SR–1504HZ/TS2000L

Industrial Robot

MAINTENANCE MANUAL

(Large-Sized SCARA Robot)

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before installing and operating the industrial robot, read through and completely understand this manual. After reading through this manual, keep it nearby for future reference.

Rev.3

April, 1 2020

Shibaura Machine

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1. Warranty period
   SHIBAURA MACHINE agrees to repair or replace as necessary all defective material or workmanship up to the period shown below, whichever comes first.
   1) Twenty-four (24) months from the date of dispatch from our plant.
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   3) 4,000 running hours from the date of initial machine operation.

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   1) Only the product delivered to the customer is subject to SHIBAURA MACHINE's Guarantee. Such Guarantee covers the specifications and functions as defined in the product specifications manual, catalog, instruction manual, etc. In no event does the Guarantee cover any secondary or associated damage caused by malfunction of the delivered product.
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   SHIBAURA MACHINE's Guarantee shall not cover the following cases.
   1) Incorrect use not described in the instruction manual, and trouble or damage caused by negligent use.
   2) Inconvenience caused by aged deterioration or long-term usage (natural fading of coating or painting, deterioration of consumable parts (*1), etc.).
   3) Inconvenience caused by sensuous phenomena (noise generation, etc. which will not affect the function).
   4) Remodeling or disassembly which SHIBAURA MACHINE will not permit.
   5) Trouble and damage caused by insufficient maintenance/inspection or improper repair.

6) Trouble and damage caused by disaster, fire or other external factor.

7) Internal data such as program and point which were created by the customer.

8) When the robot purchased in Japan was shipped overseas.

(*1) The consumable parts signify the replacement parts for maintenance as listed in Appendix Tables 1 through 6 of this manual.

4. Precautions

1) Unless the product was used pursuant to its specifications, SHIBAURA MACHINE will not guarantee the basic performance of the product.

2) If the customer did not observe the warnings and cautions described in this manual, SHIBAURA MACHINE will not assume the responsibility for any consequential accident resulting in injury or death, damage or trouble.

3) Please note that the warnings, cautions and other descriptions stipulated in this manual are only those which can be assumed by SHIBAURA MACHINE as of now.

INTRODUCTION

This manual describes the maintenance of SR series large-sized SCARA robot SR–1504HZ and controller TS2000L.
The maintenance and inspection are essential to maintain the robot performance for long years to prevent a trouble and improve the safe work.
Before starting an actual operation, it is strongly recommended to read through this manual and draw up a maintenance schedule.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will imminently lead to fatalities or serious injuries."
 **WARNING**	This means that "incorrect handling may lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

[Maintenance and inspection]

Be sure to observe the following items to use the robot safely.

 **DANGER**	
 Prohibited	• DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture.
 Mandatory	• Be sure to turn off the main power switch of the controller before starting inspection or maintenance. • Batteries should be disposed of according to the user's in-house regulations.

! CAUTION	
 Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
 Mandatory	• Always use the SHIBAURA MACHINE's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

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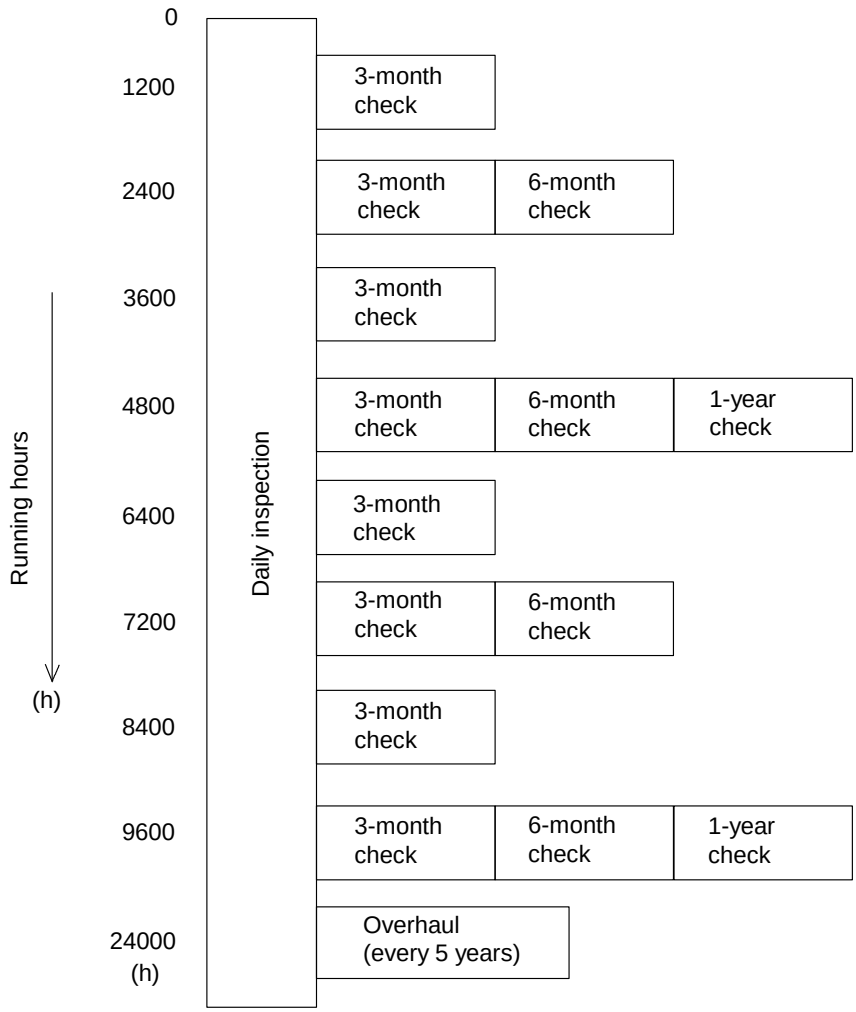
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1. Maintenance

1.1 Maintenance Schedule

Maintenance comes in the two (2) types; daily inspection, and regular inspection and maintenance. For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



Inspection schedule

- When the system is operated for 16 hours per day:
  16 hours × 25 days × 3 months = 1,200 hours
- When the system is operated for 24 hours per day:
  24 hours × 25 days × 3 months = 1,800 hours

1.2 Items for Maintenance and Inspection

This section describes the items for maintenance and inspection.
For the executing procedures, see the relevant paragraph listed in the table below.

a) Daily inspection

No.	Item	Inspection procedures	Relevant Para.
1	Tool fixture	Make sure that the clamp bolts of the tool set flange are tightened completely.	2.4.1
2	Robot installation bolts	Make sure that the robot column installation bolts are tightened completely.	2.4.2
3	Vibration and noise	Make sure that each axis can move smoothly.	
4	Change in positioning accuracy	Make sure that the robot can stop at a predetermined position accurately.	
5	Controller vent holes	(1) Make sure that no obstacle stands in front of the vent holes. (2) Make sure that no contaminant is left in the vent hole.	3.5.1
6	Function of safety devices	Make sure that each safety device connected to the emergency stop pushbutton switch and external operation input signal can function properly.	3.5.2

b) Quarterly (every three (3)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Lubricated state of linear guides and ball screw	Make sure that the linear guides and ball screw of the axis 3 are greased completely and that no rust is developed on them.	2.4.5

c) Semi-annual (every six (6)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Air leakage Piping and wiring	Check for the state of the flexible cables and air tubes in the arm bending sections and column duct.	2.4.3
2	Greasing of linear guides and ball screw	Grease the linear guides and ball screw of the axis 3.	2.4.5
3	Timing belt	(1) Make sure that the timing belt is not worn off or cracked. (2) Check the timing belt for tension.	2.4.6

d) Annual (every year) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Change of grease in each axis reduction gear	Replace the grease in each axis reduction gear.	2.4.5
2	Replacement of position detector batteries	Replace the position detector backup batteries. (They are stored in the robot column junction box.)	2.4.7

e) Overhaul (every five (5) years)

No.	Item	Inspection procedures	Relevant Para.
1	Inspection and replacement of consumable parts	Inspect the flexible cables, bearings, ball screw, linear guides, timing belt, motors and reduction gears, and replace them when necessary.	2.4.5 2.4.6
2	Battery replacement (Replace the battery every five (5) years.)	Replace the RAM (controller) backup battery. If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage has dropped, the battery alarm lamp illuminates. Replace the battery immediately then.	3.5.3
3	Replacement of switching power supply units (Replace the units every five (5) years.)	As the switching power supply unit uses an aluminum electrolytic condenser, its life is limited. Replace each switching power supply unit (DC5 V, DC24 V) for preventive maintenance. The life of the switching power supply unit differs with load current, running hours and ambient temperature. If the load current is large, the running time is long and the ambient temperature is high, the life becomes short.	3.5.4

1.3 Maintenance Tools

We recommend to use the following tools for maintaining the robot and controller.

- Screwdrivers (Phillips head screwdriver and slotted screwdriver)
- Hexagonal wrench set: M3 ~ M20
- Torque wrench set
- Water pump pliers
- Nippers
- Grease gun
- Seal tape
- Loctite adhesives (242; mid adhesive force)
- Belt tension meter

1.4 Maintenance Contract and Repair

1.4.1 Maintenance Contract

For the inspection items for maintenance and inspection to be done semi-annually and after, it is rather difficult for the user to execute, except for greasing and lubrication. We recommend that the user contact SHIBAURA MACHINE Service Department and conclude an after-sale service contract with us.

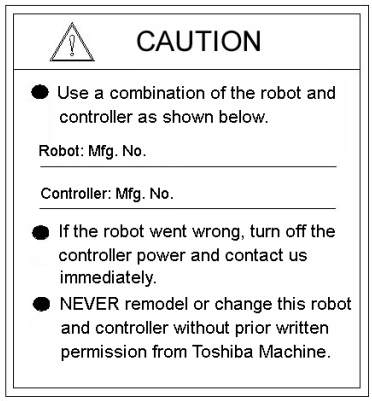
1.4.2 Repair

If a fault has occurred or if repair is necessary, turn off the controller power and contact the SHIBAURA MACHINE Service Department. At this time, advise us of the details of the fault and the following information stated on the robot and controller.

- Robot model
- Manufacture number
- Manufacture number of controller
- Date of manufacture of robot
- Running hours

For the manufacture number and date of manufacture, see the following labels attached to the column lateral side of the robot.





1.4.3 Modification

The robot and controller MUST NOT be remodeled or changed without a prior consent from SHIBAURA MACHINE.

CAUTION

- The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.

2. Maintenance of the Main Robot

2.1 Cautions on Maintenance and Inspection

When performing inspection or maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

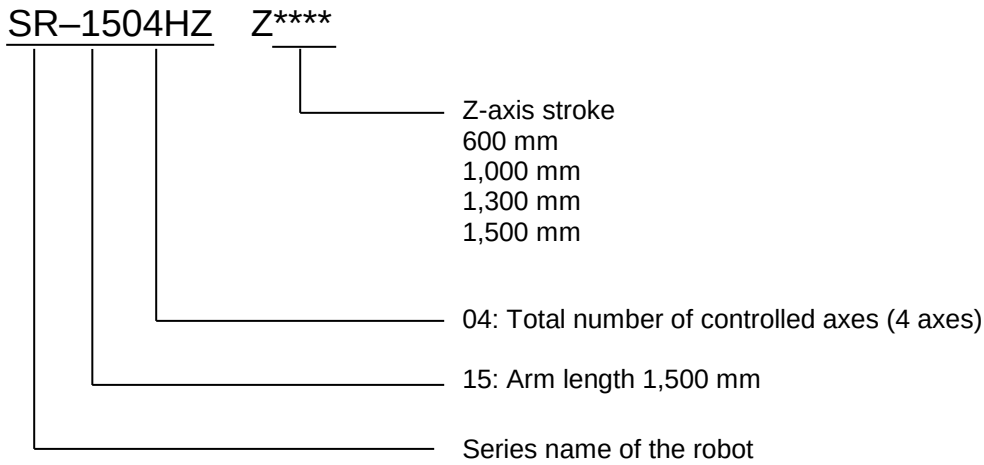
! DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

2.2 Identifying Type of SR–1504HZ

The type of the SR–1504 robot can be identified as shown below, according to the Z-axis stroke.

Example:



2.3 Layout of Robot Mechanical Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 2.1.

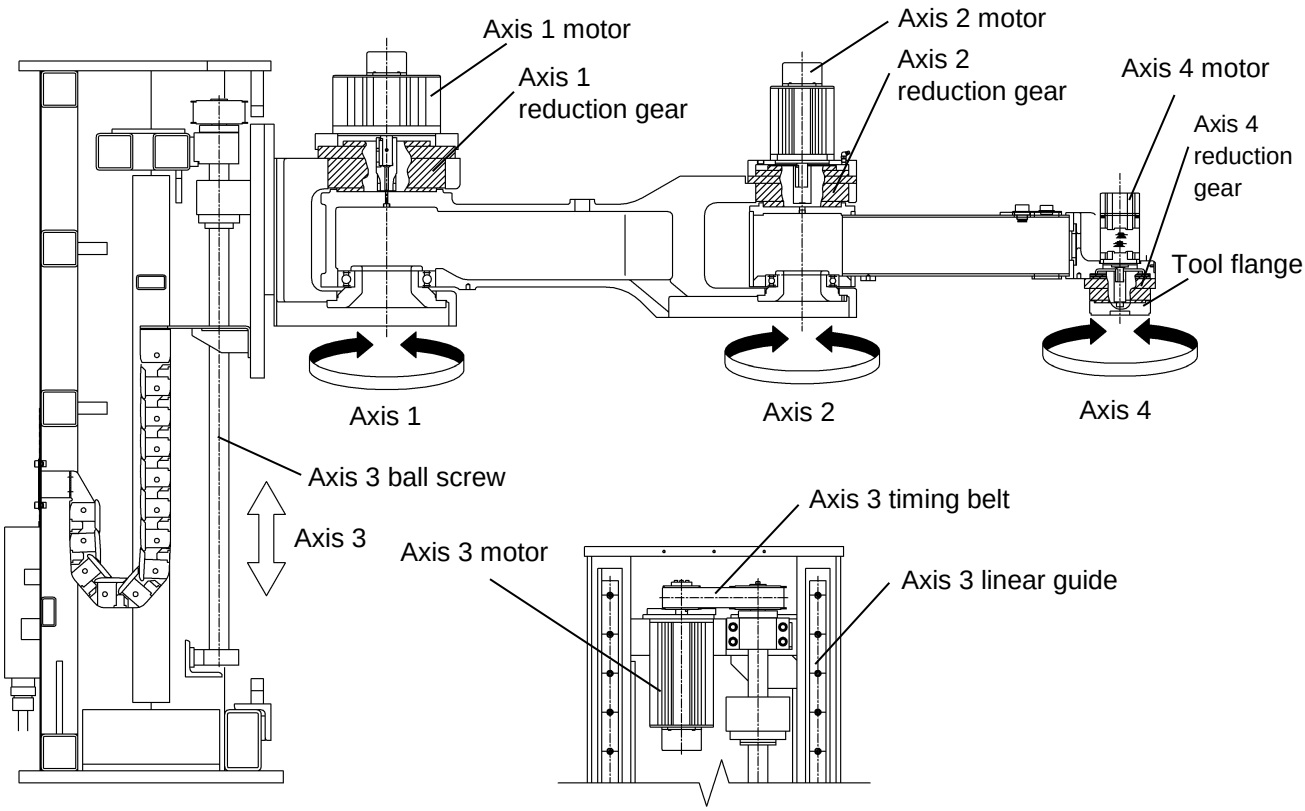


Fig. 2.1 Layout of robot mechanical components

The drive mechanism of each axis is described below.

a) Axis 1

The output of axis 1 AC servo motor is input to the reduction gear. The output of the reduction gear directly controls the arm 1 motion.

b) Axis 2

The output of axis 2 AC servo motor is input to the reduction gear. The output of the reduction gear directly controls the arm 2 motion.

c) Axis 3

The output of axis 3 AC servo motor is input to the ball screw via the timing belt to move the entire arm up and down.

d) Axis 4

The output of axis 4 AC servo motor is input to the reduction gear. The output of the reduction gear directly controls the tool flange.

2.4 Robot Maintenance Procedures

DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.
- When moving the robot by hand while the power is turned on, be sure to assure the safe work and effect an emergency stop beforehand.

2.4.1 Confirming Screws and Bolts of Each Part

Before starting an operation, make sure of the following items, using a hexagonal wrench key and driver.

a) Make sure that all bolts securing the entire robot are tightened completely.

b) Make sure that all clamp screws of each cover are tightened completely.

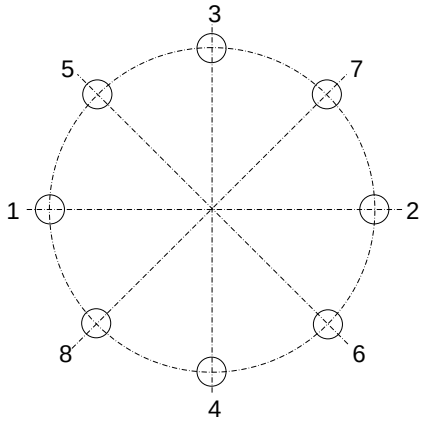
c) Make sure that all clamp bolts of the tool set flange are tightened completely.

2.4.2 Clamping Screws and Bolts of Each Part

This robot uses hexagon socket head cap screws at places requiring mechanical strength. At the time of factory-assembly, lock adhesives are applied and each screw is tightened with the following clamping torque.
When these screws are tightened again according to the inspection and maintenance as stipulated in this manual, use a torque wrench, etc. to assure the appropriate clamping torque.

Screw	M4	M5	M6	M8	M10	M12
Clamping torque	4.7 N·m	8.8 N·m	17 N·m	27 N·m	54 N·m	72 N·m

Tighten the screws arranged on a circle in the diagonal order, as shown below.



DO NOT tighten one (1) screw at a time. Tighten each screw in multiple steps, using a hexagonal wrench key, and secure with appropriate clamping torque by means of a torque wrench.

2.4.3 Confirming Wiring and Piping

Fig. 2.2 shows the actual piping and wiring drawing.
Dismount each cover and make sure that all cables and air hoses in the arm bending sections and column duct are not cracked or damaged due to twist, bend or abrasion.
If the cable or air hose is found abnormal, replace it with a new one because it may cause cable breakage, etc.

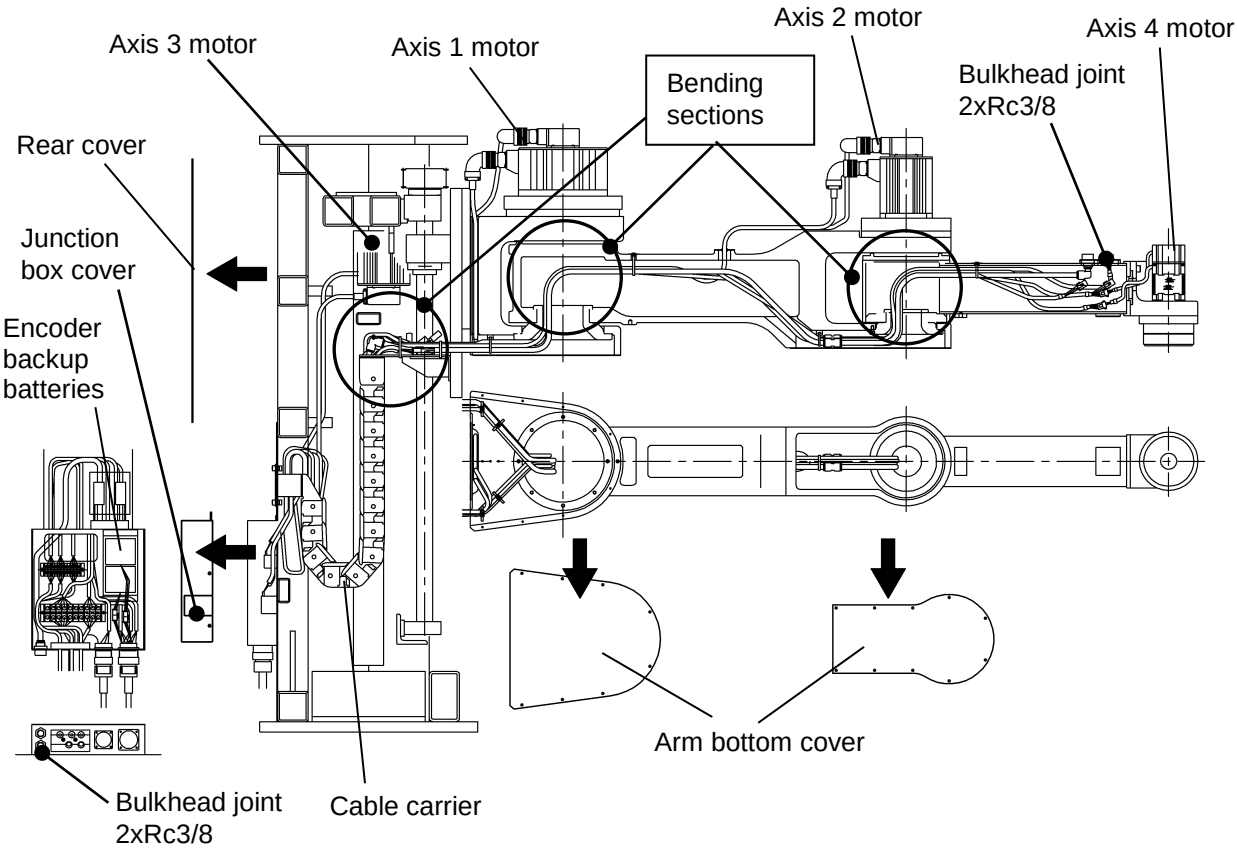


Fig. 2.2 Actual wiring and piping drawing

2.4.4 Connection Diagram

Fig. 2.3 shows the connection diagram between the devices.

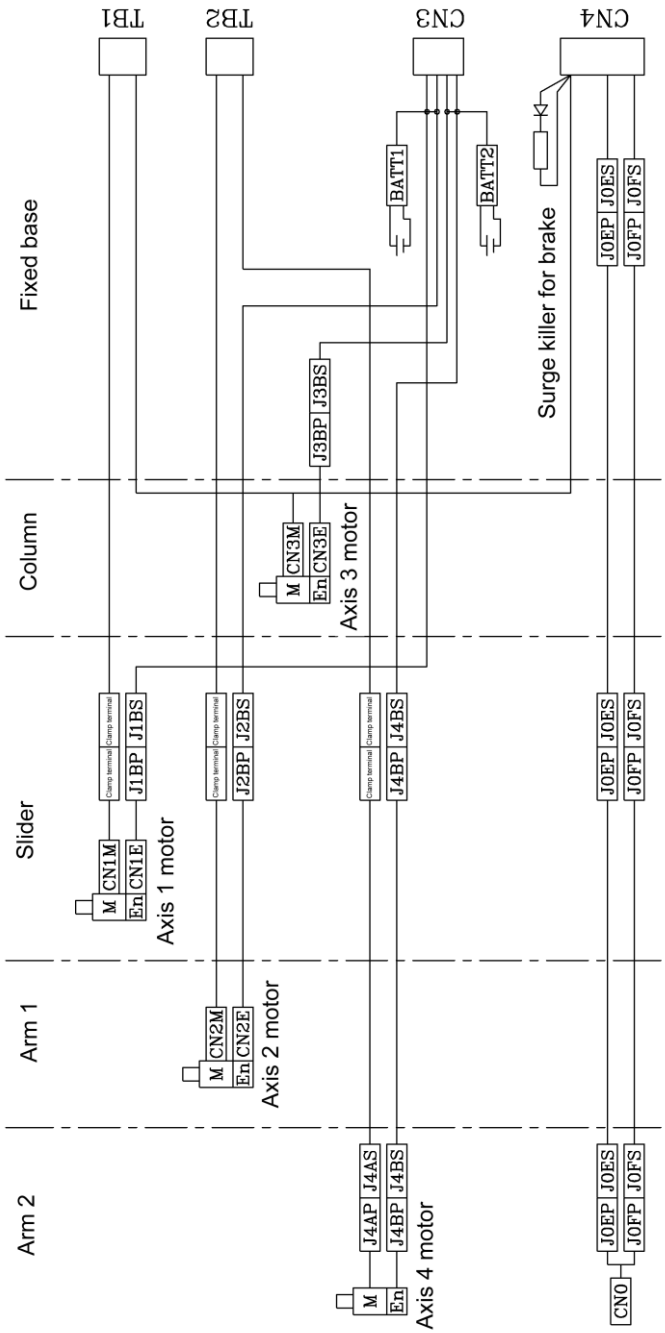


Fig. 2.3 Connection diagram between devices

2.4.5 Greasing

All mechanical sections are grease-lubricated.
Fig. 2.4 shows the grease filling points. The bearings at other than the grease-filling points indicated in Fig. 2.4 are of a grease-sealing type, and need not be filled with grease.
Two (2) types of grease are used for lubrication; grease for the reduction gear and grease for ball screw and linear guides (LM guide).

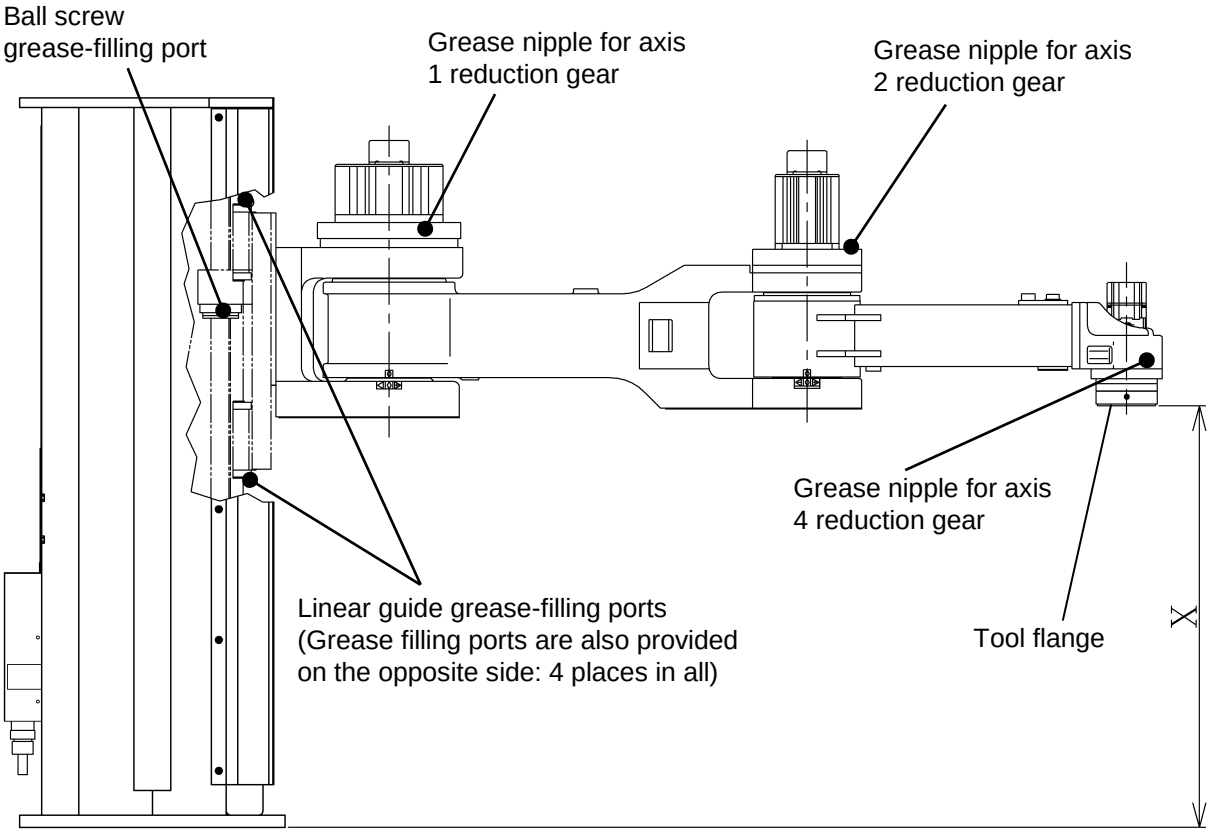


Fig. 2.4 Grease-filling points

Length “X” in the above figure signifies the tool flange position when filling grease to the linear guides, which differs with the axis 3 (Z) stroke of the robot.
For the greasing positions of each axis 3 (Z) stroke, refer to the table below.

Axis 3 stroke	Z600	Z1000	Z1300	Z1500
Length “X”	855	1090	1145	1420
Axis 3 coordinate	506	730	785	1060

a) Greasing reduction gears

Grease used for each reduction gear should be changed every 4,800 hours (approximately one (1) year).

Use the following grease for each reduction gear.

- Alvania EPR0 (maker: Shell Oil)

! CAUTION

- Take careful precautions not to mistake the type of the grease you use. Otherwise, the robot performance will drop and the robot will go wrong.

To change the grease with fresh one, remove the plug for grease drain (provided inside the bottom cover of the axes 1 and 2) and fill the grease from the grease supply port (grease nipple), using a greasing gun. As the old grease is discharged at this time, use the pipe for grease drain (standard accessory) as shown in Fig. 2.5. After the fresh grease has overflowed, change of the grease finishes.

After the greasing, cover the grease drain port with the plug again.

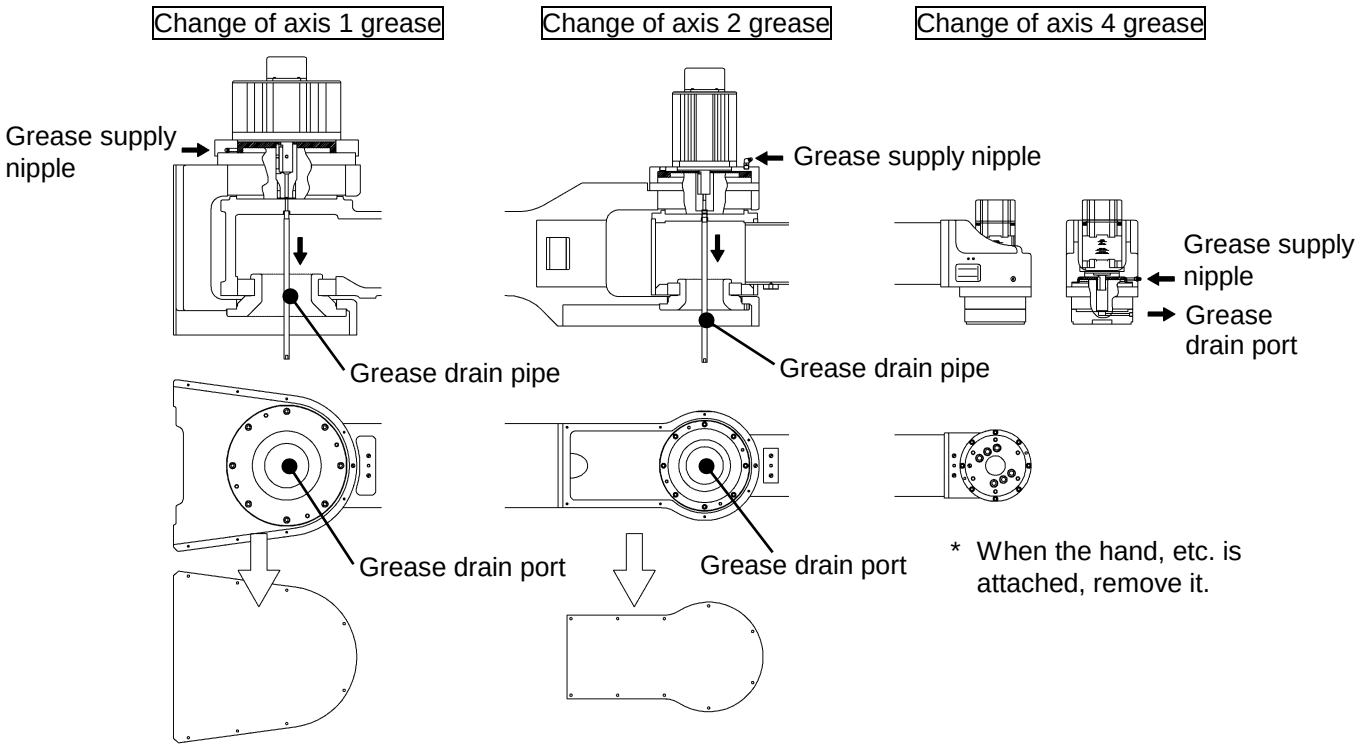


Fig. 2.5 Changing grease for axes 1, 2 and 4

The reduction gears are to be replaced at the time of overhauling. In principle, the replacement is to be done by our after-sale service engineer. If the customer changed the reduction gear, we will not guarantee any subsequent troubles or accidents.

b) Greasing ball screw and linear guides

Use the following grease for the ball screw and linear guides.

- Alvania S2 (maker: Shell Oil)

! CAUTION

- Take careful precautions not to mistake the type of the grease you use. Otherwise, the robot performance will drop and the robot will go wrong.

Greasing linear guides

To grease the linear guides, remove each cap of the grease supply port provided on each side cover of the robot. (See Fig. 2.6.) Then, guide the axis 3 toward the plus (+) direction, stop it at a position where the two (2) grease nipples of the linear guide can be seen from the grease supply port, then turn off the servo.
For the locations of the grease supply ports, see the table in Para. 2.4.5.
Fill the specified grease, using the grease gun.

Greasing ball screw

To grease the ball screw, remove the rear cover and fill the grease from the robot rear side (Fig. 2.6).

Ball screw greasing volume Linear guide greasing volume	10 cm^3

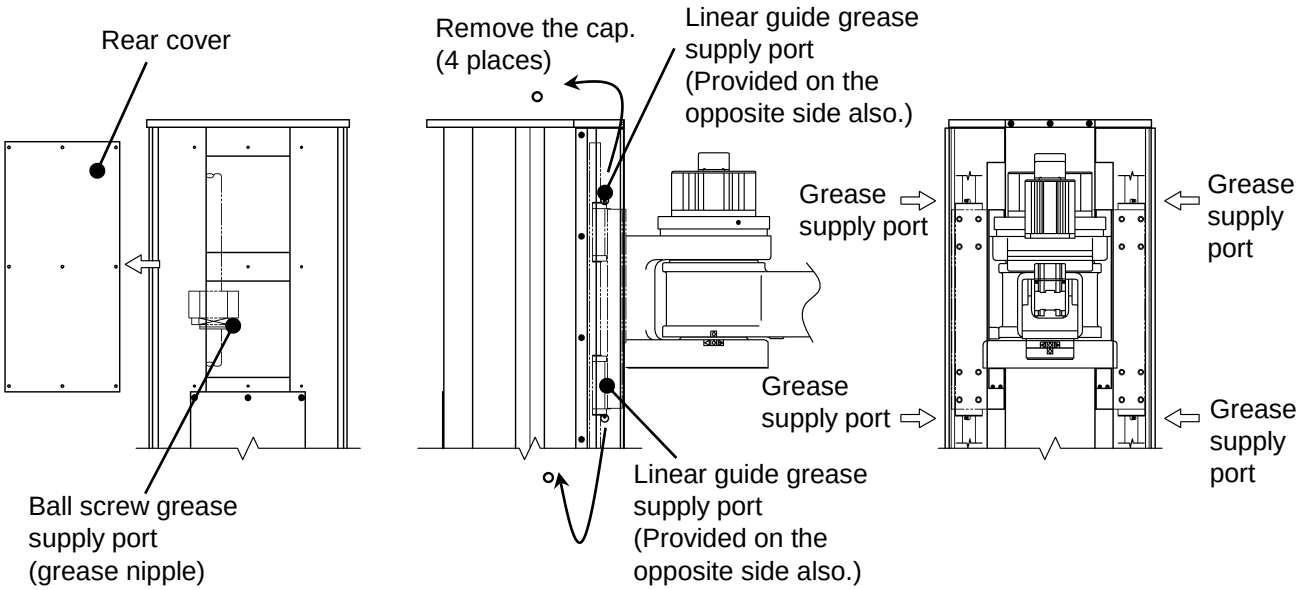


Fig. 2.6 Greasing the axis 3

The ball screw and linear guides are to be replaced at the time of overhauling. In principle, the replacement is to be done by our after-sale service engineer. If the customer changed the ball screw and linear guides, we will not guarantee any subsequent troubles or accidents.

2.4.6 Confirming Timing Belt Condition

Remove the column top cover of the robot and make sure visually that the axis 3 belt is not worn off or cracked. Also check the timing belt for the tension and adjust if loosened. Adjust the tension so that when the center of the axis 3 timing belt is pressed with the force of about 38 N (approximately 3.9 kgf), the belt deflection can be about 3 mm (= (d1 + d2) / 2). (See Fig. 2.7.)
To adjust the belt tension, use the belt tension adjusting tool attached to the robot column. Also, when the belt tension meter (such as U–507 made by Gates Unitta Asia) is provided, substitute the parameter values (belt width, span, belt mass) in Table 2.1 below, then re-adjust the set tension.

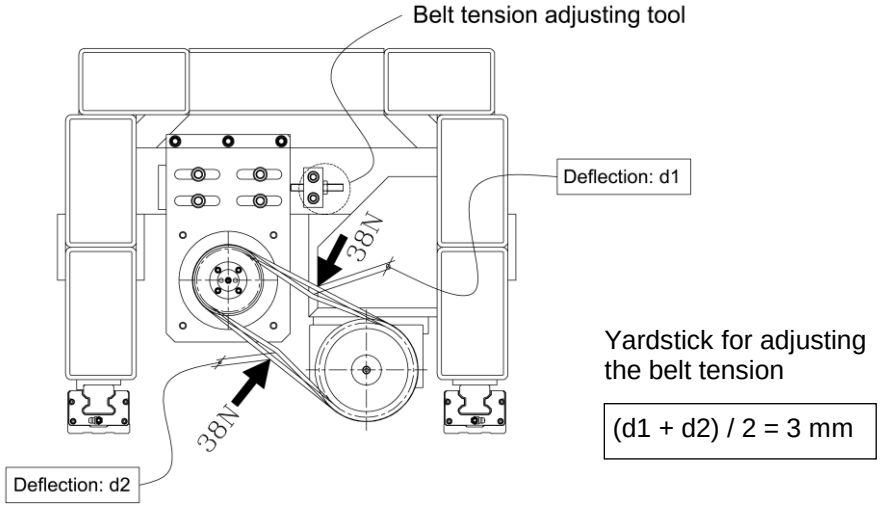


Fig. 2.7 How to confirm axis 3 belt tension

Belt type	Set tension [N]	Belt width W [mm]	Span S [mm]	Belt mass M [g/mm W×m L)
656–8YU–40 (*1)	420 ~ 480	40	184.4	5.2
632–EV8YU–40 (*2)	420 ~ 480	40	184.4	5.1

*1: Used for the machines delivered before July, 2007.
*2: Used for the machines delivered in July, 2007 and after.

Table 2.1 Substitution values for belt tension meter

In principle, the timing belt is to be replaced by our after-sale service engineer. If the customer changed the timing belt, we will not guarantee any subsequent troubles or accidents.

DANGER

- The axis 3 motor is provided with a brake. If the axis 3 timing belt is removed, the brake will not work and the axis 3 (i.e., Z-axis) will drop due to self-weight of the arm, which is very dangerous.
- If the timing belt is removed, the mechanical home point will shift and correct control is not possible.

CAUTION

- The belt tension is adjusted at a high place. Take careful precautions on the safety.
- Be careful not to cut or scratch the belt.

2.4.7 Replacing Position Detector (or Encoder) Batteries

To back up the data of the position detector attached to each motor, batteries (alkaline dry batteries) are used.
Though the battery life will vary with the controller power OFF time, be sure to replace the batteries with new ones at the time of annual inspection. When the robot is not used for a long term (i.e., the batteries are left intact in the power OFF condition), replace the batteries at the time of robot startup. As a yardstick, the shutdown period is two (2) months.
The battery replacement procedures are given below.

<Replacement procedures>
When replacing the batteries after the controller power is turned on for one (1) hour or over, start with Step 2). Otherwise (i.e., when replacing the batteries after the controller power is turned on within one (1) hour), begin with Step 1).

1) Turn on the controller power for one (1) hour or over.
2) Turn off the controller power.
3) Remove the junction box cover of the main robot.
4) Remove the battery box cover set inside, then the batteries.
5) Set the new batteries (six (6) pcs. in all) within five (5) minutes, taking careful precautions not to mistake the "+" and "–" polarities.
6) Set the battery box cover.
7) Set the junction box cover of the main robot.

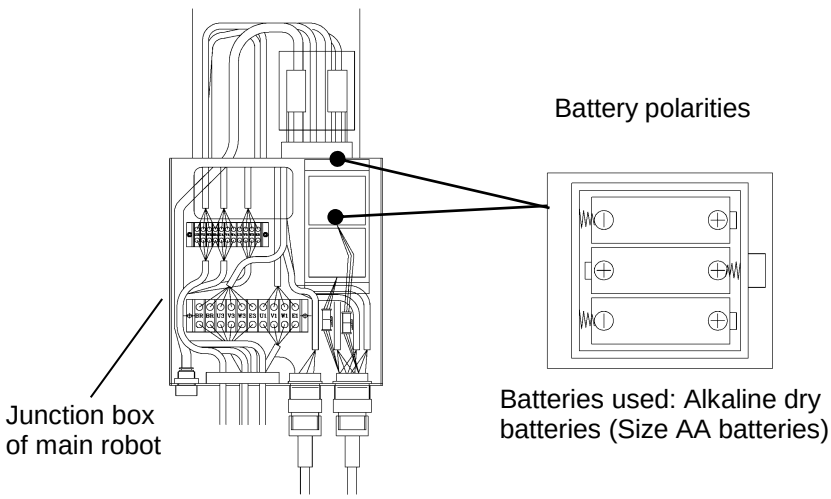


Fig. 2.8 Position detector (encoder) batteries set positions

! CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

Cautions:

If the battery voltage has dropped, either of the following battery alarms will occur.

1–049 Axis1 Battery alarm
1–065 Axis2 Battery alarm
1–081 Axis3 Battery alarm
1–097 Axis4 Battery alarm
1–113 Axis5 Battery alarm

When the batteries have been replaced just after the battery alarm generation, the battery voltage returns to normal and relevant battery alarm is reset automatically. Otherwise, the battery voltage drops further with either of the following position detection error generated.

8–065 Axis1 Encoder abnormal
8–097 Axis2 Encoder abnormal
8–129 Axis3 Encoder abnormal
8–161 Axis4 Encoder abnormal
8–193 Axis5 Encoder abnormal

The above errors signify that the position data detected by each encoder is not reliable any further. When this happens, the robot enters an emergency stop state to prevent the servo from turning on. Also, if the power is turned off in this condition, the position data is lost.
If the “axis encoder abnormal” error has generated, restore the system, referring to Para. 2.4.11 below.

2.4.8 Restoring Home Position

Batteries for backing up the position information of the position detector are housed in the junction box of the main robot. (See Fig. 2.8.)
Unless the robot has been used for a long term, the battery voltage drops and the home position data will be lost. Also, when the battery voltage for backing up the internal memory of the X8HC printed board in the controller has dropped, user's set programs, parameters and absolute coordinates will be lost. To avoid this, replace the batteries on a regular basis, taking the following information as a yardstick.

Position detector batteries:	One year after delivery (when the working time is sixteen (16) hours per day).
X8HC board SRAM battery:	Five (5) years after delivery.

The following paragraphs deal with the home position restoring procedures after the home position data has been destroyed due to battery voltage drop, etc.

2.4.9 Restoring Home Position Data Using HOME Function

<Outline of HOME function>

This function memorizes a specific coordinate value in each axis and restores the original home position data by using the HOME function when the machine home position data has been destroyed.

The set points are provided for HOME1 to HOME4, and the axis 1 to axis 3 plus (+) side mechanical stopper positions are set in HOME3, and the axis 1 to axis 3 minus (–) mechanical stopper positions in HOME4. The axis 4 is set at the 0° position according to the home point match-mark.
For HOME1 and HOME2, you can specify any values. It is recommended that when building up a system, you specify the place for securing the robot end.

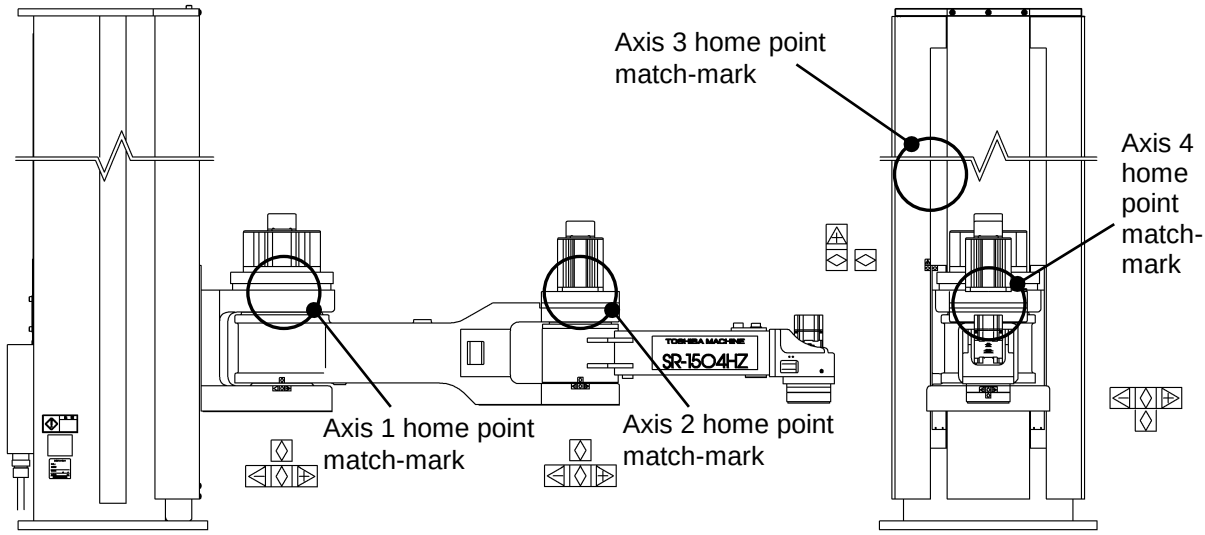


Fig. 2.9 Locations of home point match-marks

(Page 1)

USER	HOME 1	HOME 2	
(J 1)	0	0	[deg]
(J 2)	0	0	[deg]
(J 3)	0	0	[mm]
(J 4)	0	0	[deg]
(J 5)	0	0	[mm]

SET TEACH

(Page 2)

SYSTEM	HOME 3	HOME 4	
(J 1)	**1183468**	– 1180474	[deg]
(J 2)	1450716	– 1450720	[deg]
(J 3)	1239092	– 36711	[mm]
(J 4)	0	0	[deg]
(J 5)	0	0	[mm]

SET TEACH

Caution: This function is used to restore the machine home point of each axis. NEVER use the function at other than the setting for restoration.

(1) How to set data in HOME

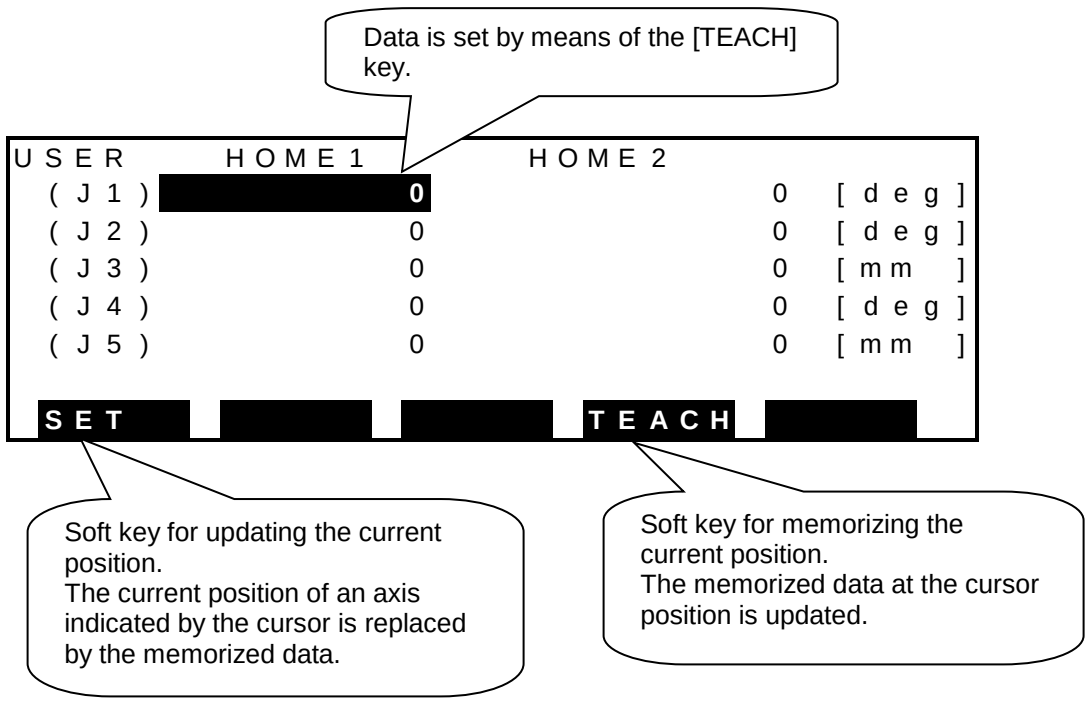
This paragraph describes the procedures for setting the data in HOME1. (The same procedures are also applicable to HOME2.)

1) Guide the robot to secure the tool flange at a desired position.

2) Turn off the servo power.

3) Press the “UTILITY” key equipped on the teach pendant.

4) Press the [NEXT] key until the [REORG] menu appears, then press the [REORG] key.



5) When the above screen appears, memorize the position where the robot was secured for each axis by moving the cursor and using the [TEACH] mode.

(2) How to restore data by HOME1

If the coordinate data of the machine home position has been destroyed, the machine home position can be restored, using HOME1 above.

1) Guide the robot and secure it at the position set in HOME1 above.

2) Turn off the servo power.

3) Press the “UTILITY” key equipped on the teach pendant.

4) Press the [NEXT] key until the [REORG] menu appears, then press the [REORG] key.

5) Move the cursor to each axis under HOME1 of the REORG screen, then press the [SET] key to update the coordinate value set in HOME1 to the position where the robot was secured. Thus, the machine home position can be restored. When setting the axis 3 data at this time, set the axis 4 data beforehand.

(3) How to restore data by HOME3 (HOME4)

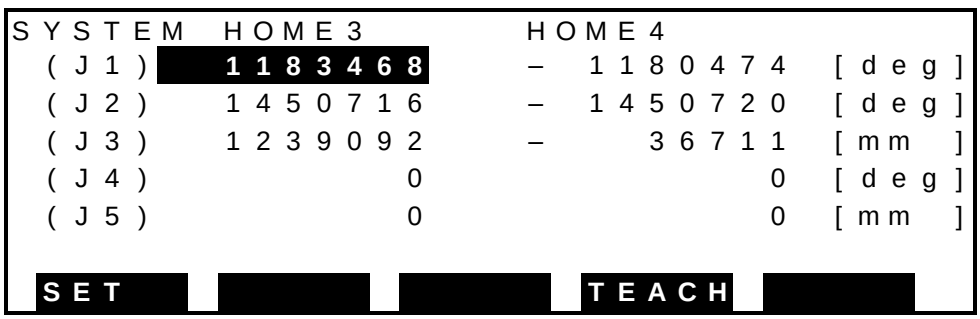
The plus (+) side mechanical stopper positions and minus (–) side mechanical stopper positions of the axes 1 to 3 are factory-set in HOME3 and HOME4, respectively. The axis 4 is set to the home point match-mark.
The machine home point can be restored, using the above.
This paragraph deals with the restoration procedures, using HOME3.

1) Turn off the servo power.

2) Press the “UTILITY” key provided on the teach pendant.

3) Press the [NEXT] key until the [REORG] menu appears, then press the [REORG] key.

4) Press the [NEXT] key to call Page 2.



5) Contact the axis 1 of the robot to the plus (+) side mechanical stopper by hand.

6) Move the cursor to (J1) of HOME3, then press the [SET] key. Now, the machine home point of the axis 1 is restored.

7) Likewise, contact the axis 2 of the robot to the plus (+) side mechanical stopper by hand.

8) Move the cursor to (J2) of HOME3, then press the [SET] key. Now, the machine home point of the axis 2 is restored.

9) Press the [ESC] key to escape from the REORG screen once.

10) Turn on the servo power, and set the axes 3 and 4 in the servo-free state.

11) Set the axis 4 to the home point match-mark.

12) Contact the axis 3 of the robot to the plus (+) side mechanical stopper by hand.

13) Turn off the servo power.

14) Repeat Steps 2) to 4) inclusive to call the HOME screen again.

15) Move the cursor to (J4) of HOME3, then press the [SET] key. Now, the machine home point of the axis 4 is restored.

16) Move the cursor to (J3) of HOME3, then press the [SET] key. Now, the machine home point of the axis 3 is restored.
Press the [ESC] key to escape from the HOME screen. Now, the operation has finished.

2.4.10 Restoring Home Position Data, Using Home Point Setting Tool

This paragraph describes the procedures for restoring the home position data, using the home point setting tool (standard accessories). Provide the tool according to Appendix Table 4.

(1) How to restore axes 1, 2 and 4

1) Turn on the controller power, then manually set the axes 1, 2 and 4 match-marks while the servo power is turned off. (For the locations of each axis home point match-marks, see Fig. 2.9.)

2) Dismantle the joint bottom covers of the axes 1 and 2. When the axis 4 is equipped with a hand, etc., disconnect it.

3) Attach the home point setting tool commonly used for the axes 1, 2 and 4 to the home point setting tool set surface of the axis you wish to restore. Use the locate pins (two (2) places) of the tool and secure with the hexagon socket head cap screw (M6×120).

4) Insert the locate pin (S655130) into the locate pin insertion hole (8 mm-dia.) provided on the matching part (joint or tool flange). If the pin hole is shifted, fine-adjust it manually.

5) Perform home point setting (ZEROP) operation at a position where the home point setting tool was mounted. For details, see Para. 2.4.11.

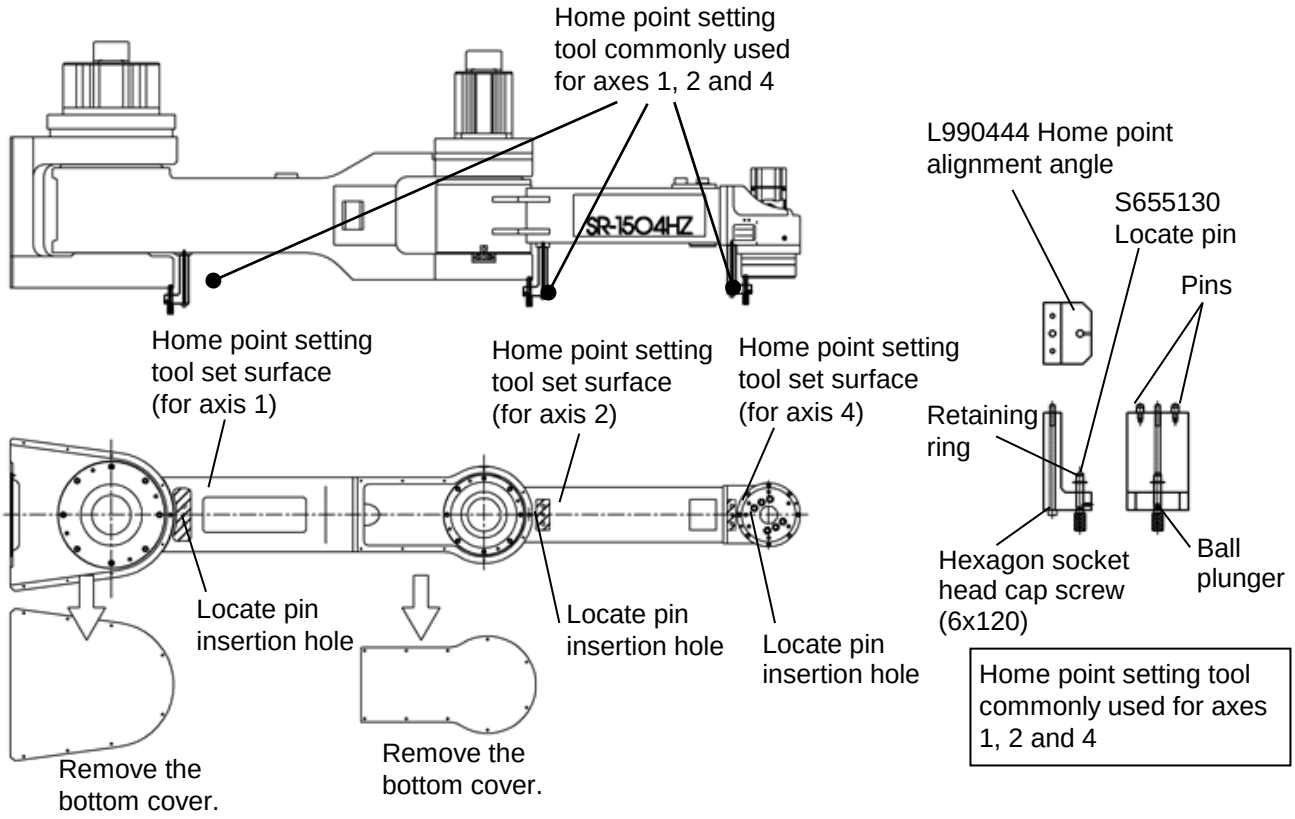


Fig. 2.10 Home point setting tool commonly used for axes 1, 2 and 4

(2) How to restore axis 3

1) Remove the front cover of the robot.

2) Guide the axis 3 (i.e., Z-axis) so that the distance between the axis 1 joint top surface and robot column top surface can fall under the vicinity of 275 mm. (See Fig. 2.11.)

3) Secure the axis 3 home point setting tool to the column top surface (i.e., ceiling plate) with the two (2) hexagon socket head cap screws (M12×50). Insert the locate pin (S655131) into the long hole provided on the Z-axis slider. If the hole position on the same slider is shifted at this time, fine-adjust it.

4) Perform home point setting (ZEROP) operation at this position. The coordinate value of the axis 3 (Z-axis) when the tool was secured will not become zero (0), however. You have to perform home point setting (ZEROP) operation based on the coordinate value assigned for each Z stroke. For details, see Para. 2.4.11.

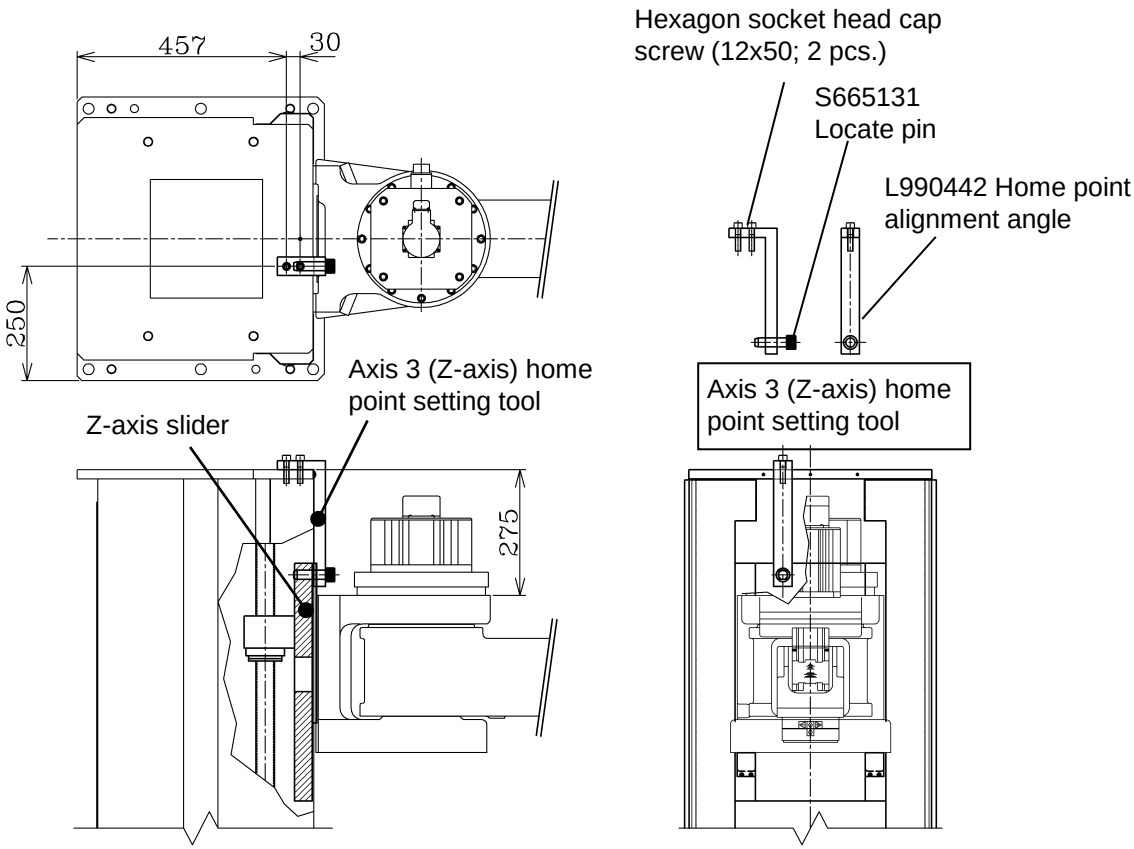


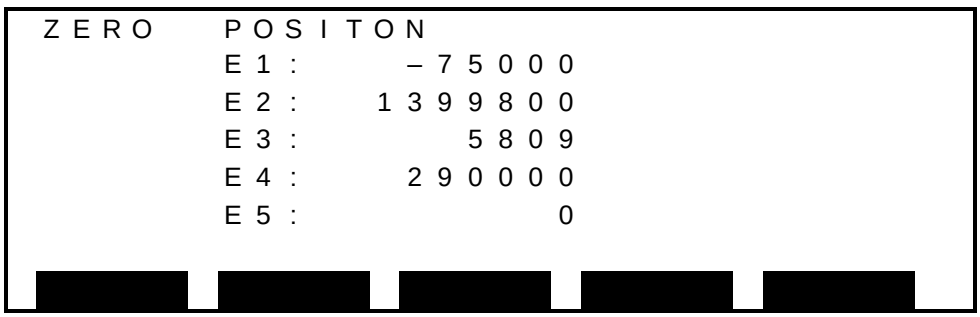
Fig. 2.11 Axis 3 (Z-axis) home point setting tool

2.4.11 Home Point Setting (ZEROP) Procedures

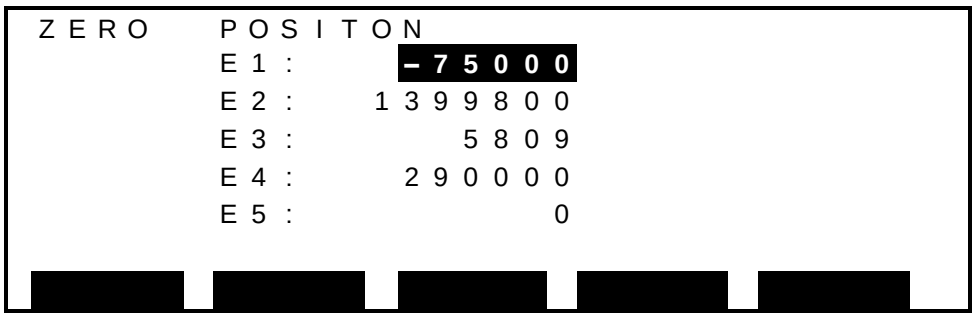
Set each axis home position, then execute the following home point setting (ZEROP) operation.

(1) Setting new home position data (through teach pendant)

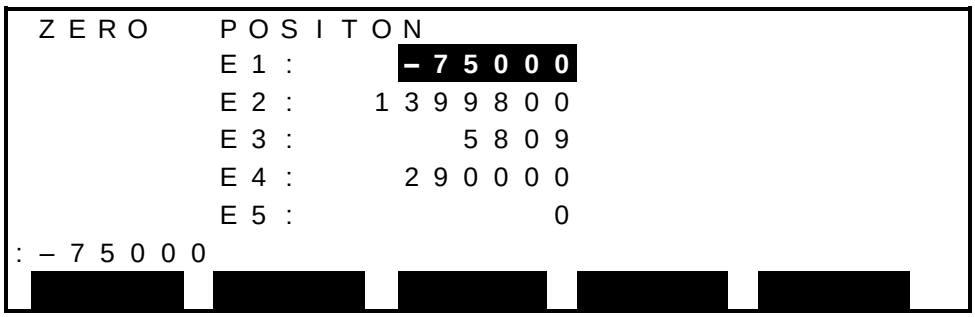
1) Select the TEACHING mode by means of the master mode switch.
2) Press the “UTILITY” key equipped on the teach pendant.
3) Press the [NEXT] (F6) key on the teach pendant twice.
4) Turn off the servo power.
5) Press the [F3] key on the teach pendant to select the ZEROP mode.
6) The following home point setting screen is called..



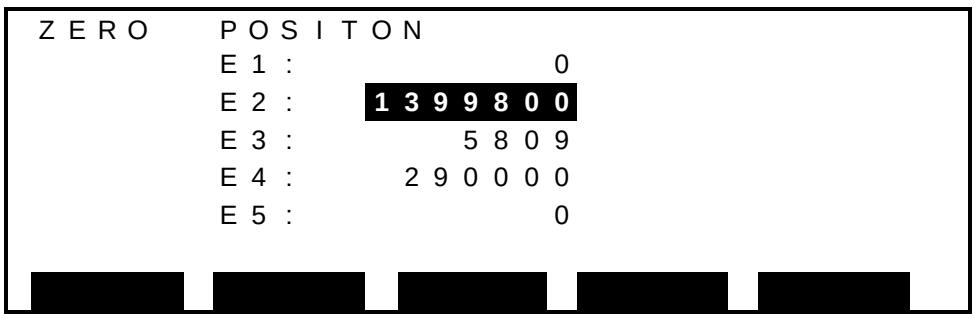
7) Press the “ALT” and “0” keys on the teach pendant at the same time.
8) The following home point editing screen appears.
9) The numerical value of E1 (axis 1) is inversely displayed.



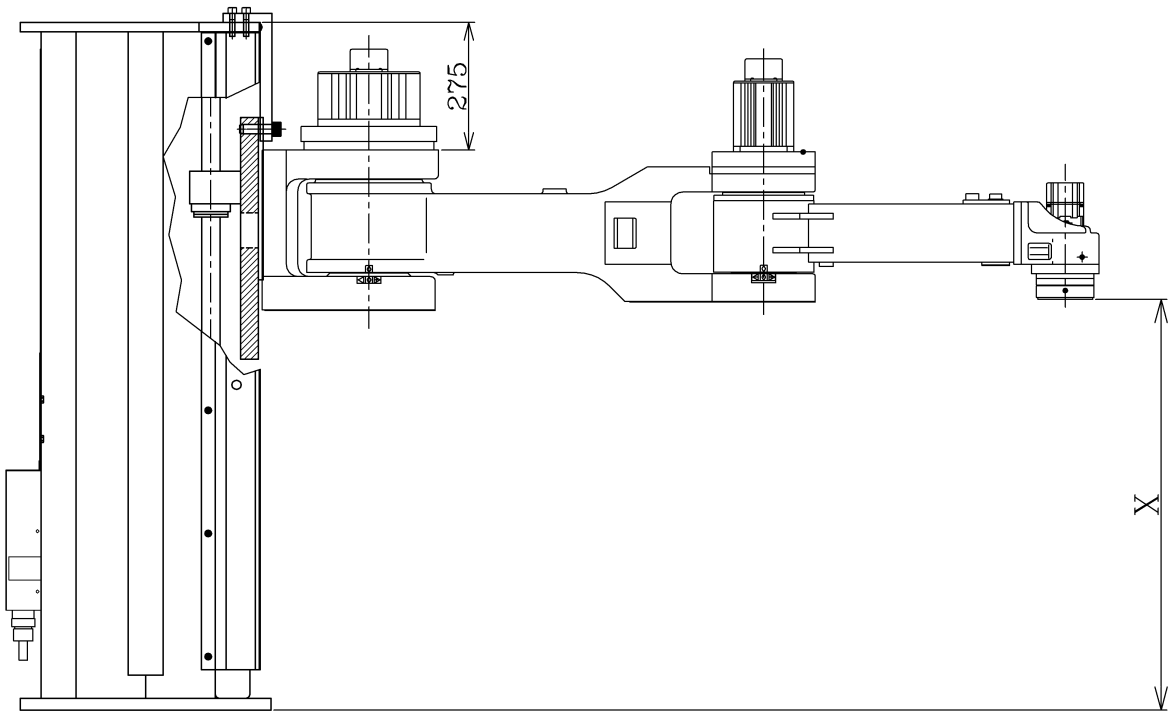
10) Press the “EXE” key on the teach pendant, and the value of “E1” is displayed at the lower left side. Set “0” in lieu of this value and press the “EXE” key. Then the E1 value becomes “0”. Unless the axis could be located at the home position, enter the coordinate value registered beforehand.



11) When the “↓” key on the teach pendant is pressed, the following E2 value is displayed by reversed image. Perform setting in the same manner as above.



12) The home point value of E3 (axis 3) which has been set, using the home point setting tool, differs with the type of Z-axis stroke. Substitute the ZEROP set value as shown in the table below, according to the type of Z stroke you are using now.



Type of Z-axis (axis 3) stroke	Tool flange height (mm) [Length “X” in the above figure]	Axis 3 coordinate value (mm)	ZEROP set value
Z600	885	525	5250000
Z1000	1285	925	9250000
Z1300	1585	1225	12250000
Z1500	1785	1425	14250000

(2) Confirming home position

1) Make sure of the robot current position according to the POS data of the UTILITY mode. Guide the robot to the approximate home position by manual operation and confirm the position.

2) Execute the following program and make sure of the robot machine home position.

```
PROGRAM ZERO
  SPEED=30
  MOVEA 3,100
  MOVEA 1,0
  MOVEA 2,0
  MOVEA 3,0
  MOVEA 4,0
END
```

(3) Saving data

Save the home position data.

Connect with the controller the personal computer where the TSPC software (option) is already installed, then save the “MACHINE.PAR” file from the robot controller to the hard disk, etc. of the personal computer.

Load the home position data to the system parameter file attached to the “MACHINE.PAR” file.

The home position data are characteristic of each robot, which differ with the robot you use. Be sure to back up the data.

Now, the operation completes.

3. Maintenance of Controller

3.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of the controller, follow the items given below so that the work can be carried out safely.

! CAUTION

- Before removing the controller cover for maintenance or inspection, be sure to turn off the main power switch of the controller.
  Also, turn off the power breaker when connecting or disconnecting the power cable.
- DO NOT touch the servo driver for at least two (2) minutes after the main power switch is turned off. The large-capacity capacitor in the servo driver may be charged with voltage, and touching the servo driver could lead to electric shocks.
- DO NOT disconnect the battery connector at other than battery replacement. Otherwise, files saved in the memory may be lost.

! DANGER

Be sure to turn off the main power switch. If the main power switch remains ON, the servo power printed board, servo printed board and switching power supply are charged, and touching them could lead to electric shocks. To avoid this, be sure to turn off the main power switch before starting the work.

3.2 External View of Controller

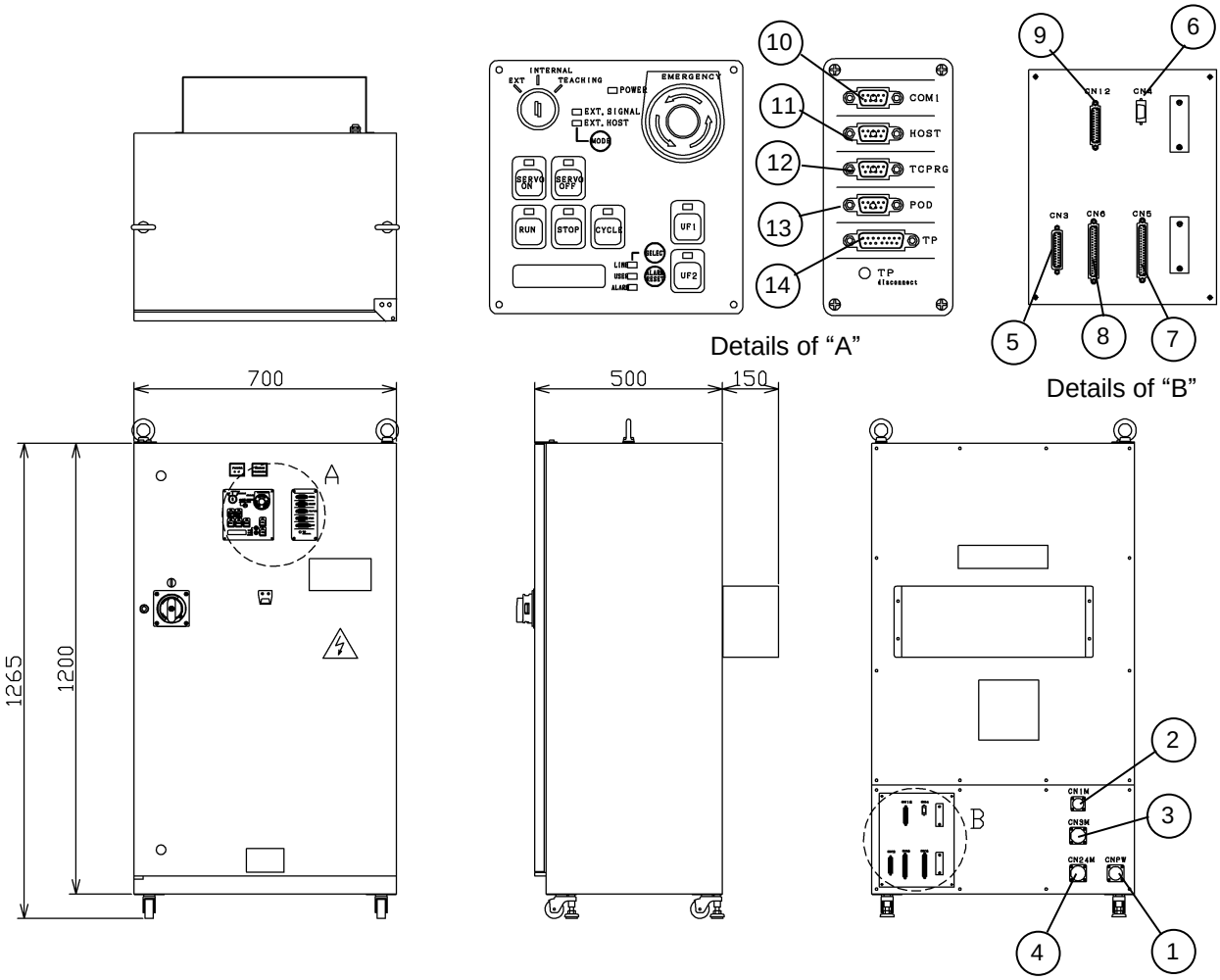


Fig. 3.1 External view of controller

No.	Part name	Description
1	CNPW	Power supply
2	CN1M	Axis 1 motor drive cable
3	CN3M	Axis 3 motor drive cable
4	CN24M	Axes 2, 4 motor drive cables
5	CN3	Robot encoder cable (for axis 1 to axis 4)
6	CN4	Hand control signal cable
7	CN5	External control input signal cable
8	CN6	External control output signal cable
9	CN12	External I/O signal cable
10	COM1	Serial communication port for external equipment
11	HOST	Serial communication port for user
12	TCPRG	Port for editing sequence program
13	POD	Port for touch panel
14	TP	Serial port exclusively used for teach pendant

3.3 Parts Layout in Controller

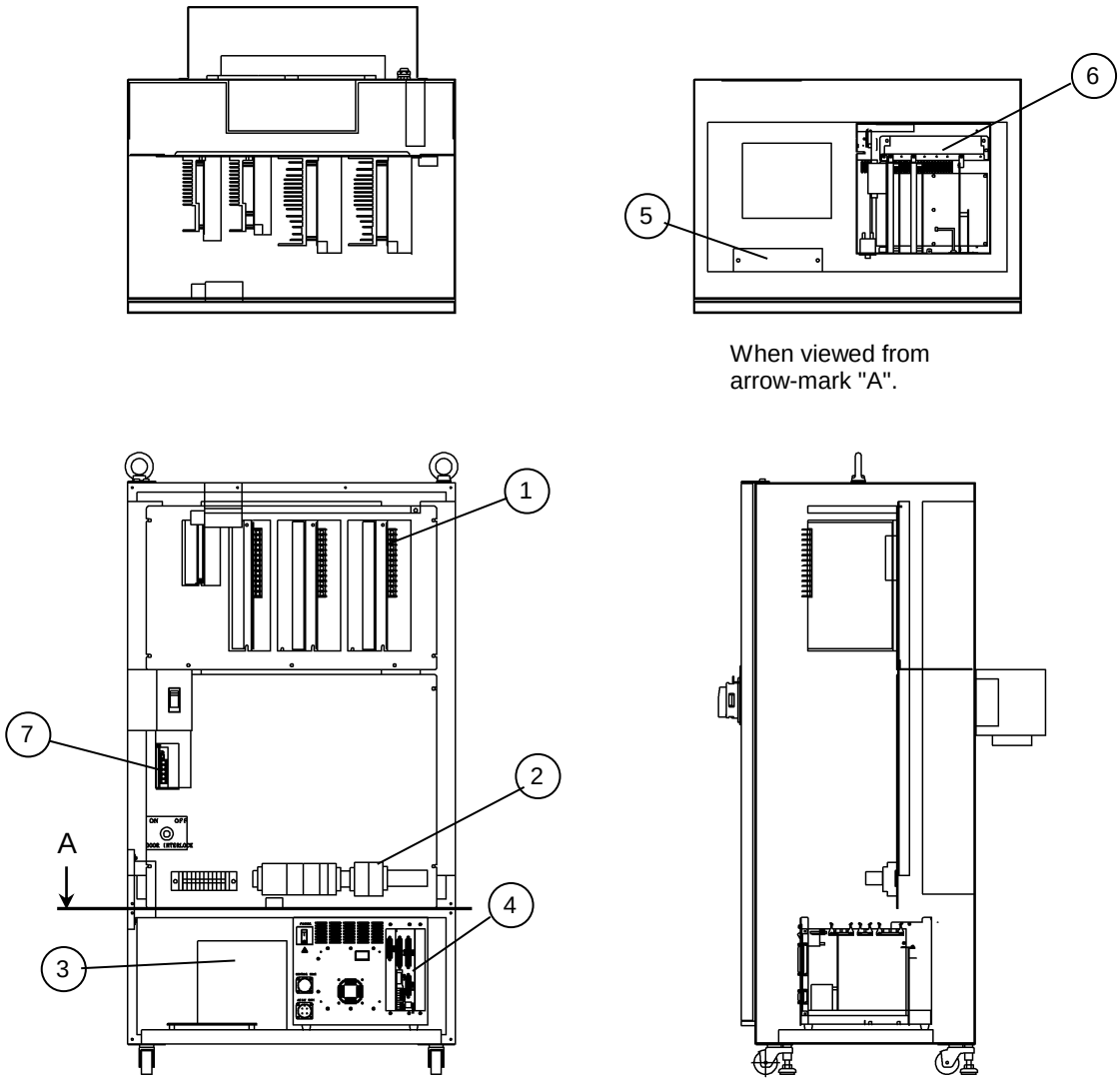
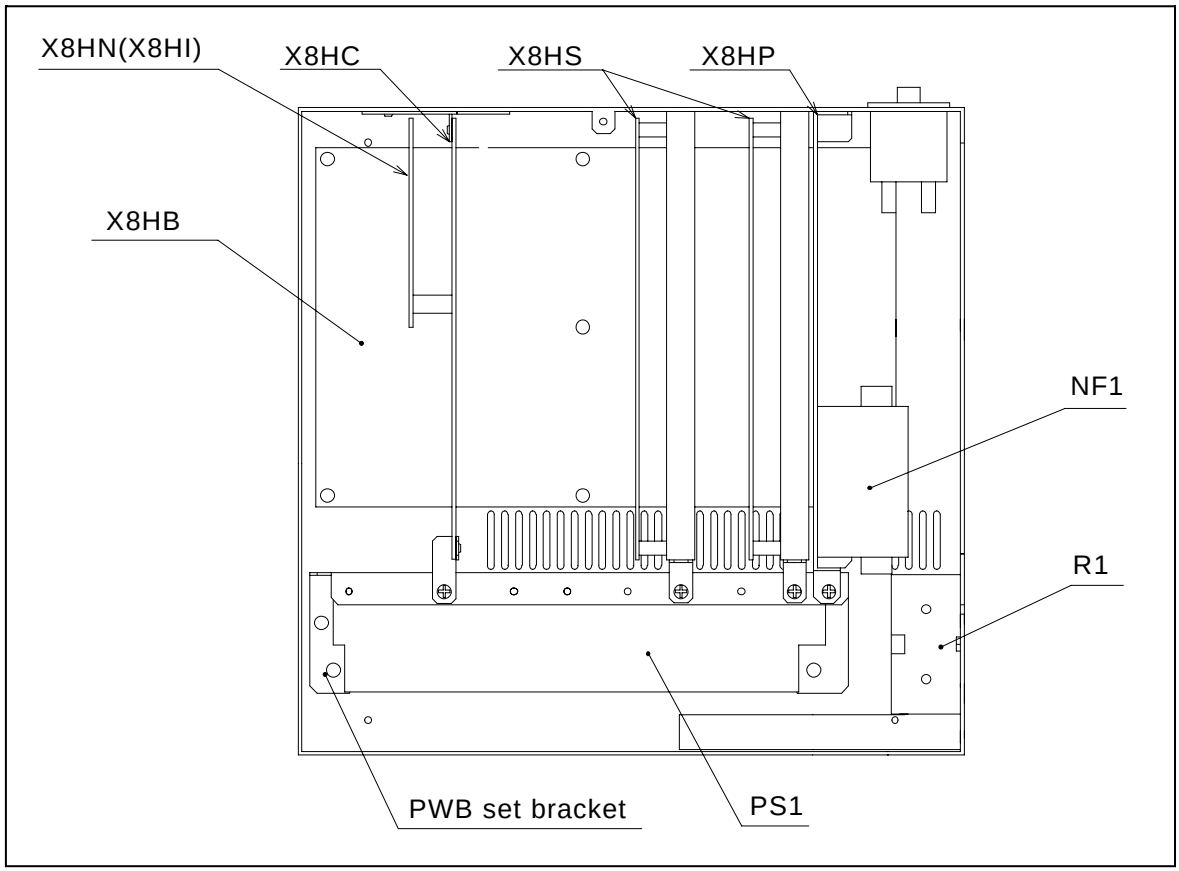


Fig. 3.2 Parts layout in controller

No.	Description
1	Servo unit
2	Circuit protector
3	Transformer
4	Controller unit (TS2000)
5	Grounding plate
6	Switching power supply PS1
7	Switching power supply GS1

3.4 Layout of Controller Components



Upper side view

Fig. 3.3 Layout of controller components

Part name	Description
PS1	P5V, P24V switching power supply
PWB set bracket	Bracket for mounting PS1
X8HP	Servo power printed board
X8HC	Main control printed board
X8HN (X8HI)	I/O output printed board
X8HS	Servo printed board
X8HB	Mother board
R1	Regenerative resistor
NF1	Noise filter

3.5 Maintenance Procedures

3.5.1 Check of Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction. To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.

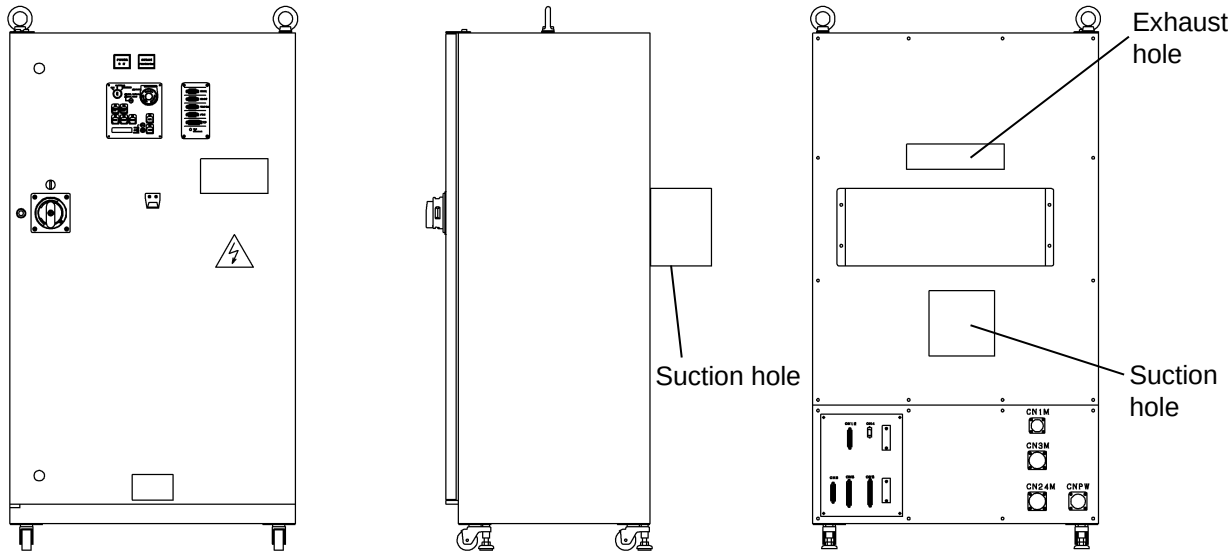


Fig. 3.4 Controller side views

a) Make sure that no obstacle stands in front of each air vent hole.
If there is an obstacle, move it away from the air vent hole so that air flow is not blocked.

b) Make sure that no contaminant is left on the air vent holes. If any contaminant is left on the vent hole, remove it.

3.5.2 Check of Safety Devices for Function

Make sure that the EMERGENCY stop pushbutton switches equipped on the control panel and teach pendant work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

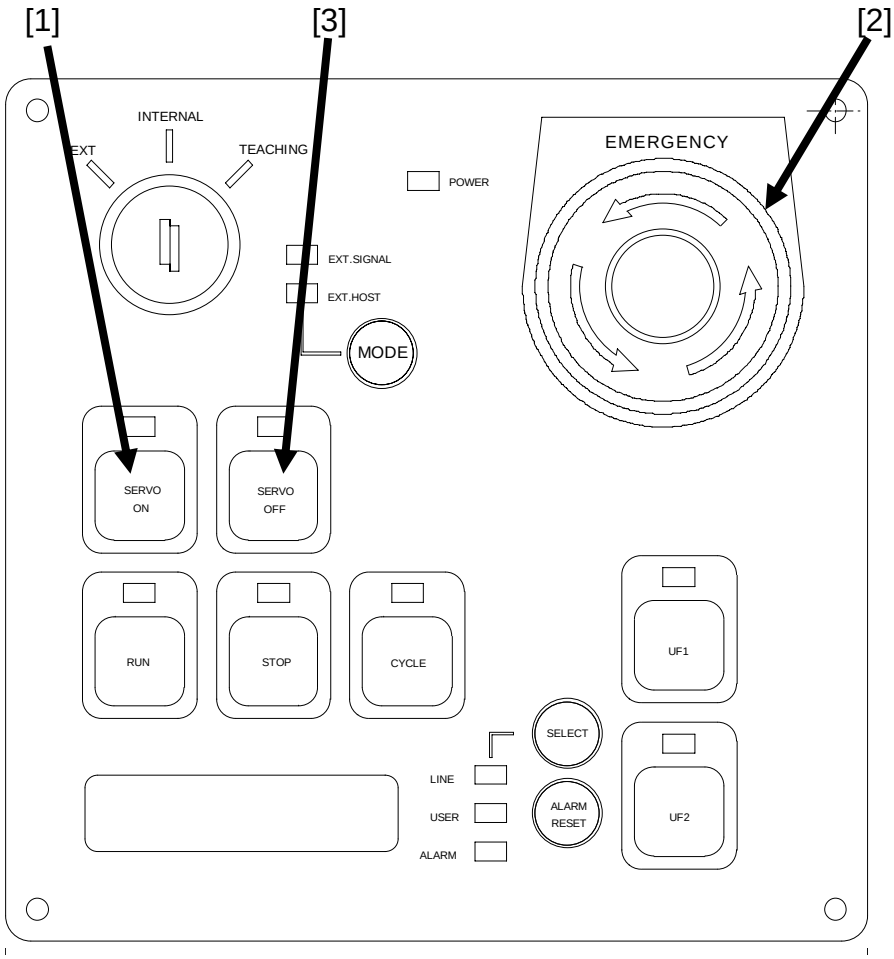


Fig. 3.5 Layout of control panel switches

a) Make sure that the EMERGENCY (emergency stop) switch [2] provided on the control panel functions properly.

1) Press the SERVO ON switch [1] provided on the control panel to turn the servos on and make sure that the SERVO ON switch [1] LED is illuminated.

2) Press the EMERGENCY switch [2] and make sure that the servos are turned off. The SERVO ON switch [1] LED goes out and the SERVO OFF switch [3] LED is illuminated. At this time, make sure that the EMERGENCY switch [2] remains depressed.

3) Turn the EMERGENCY switch [2] clockwise and make sure that the same switch turns off.

b) Make sure that the EMERGENCY STOP switch [6] provided on the teach pendant functions properly.

1) Press the SERVO ON switch [4] on the teach pendant to turn on the servo power, and make sure that the SERVO ON switch LED [5] is illuminated.

Note: The servo power cannot be turned on in the TEST RUN mode.

2) Press the EMERGENCY STOP switch [6] on the teach pendant and make sure that the servos are turned off. The SERVO ON switch LED [5] goes out.

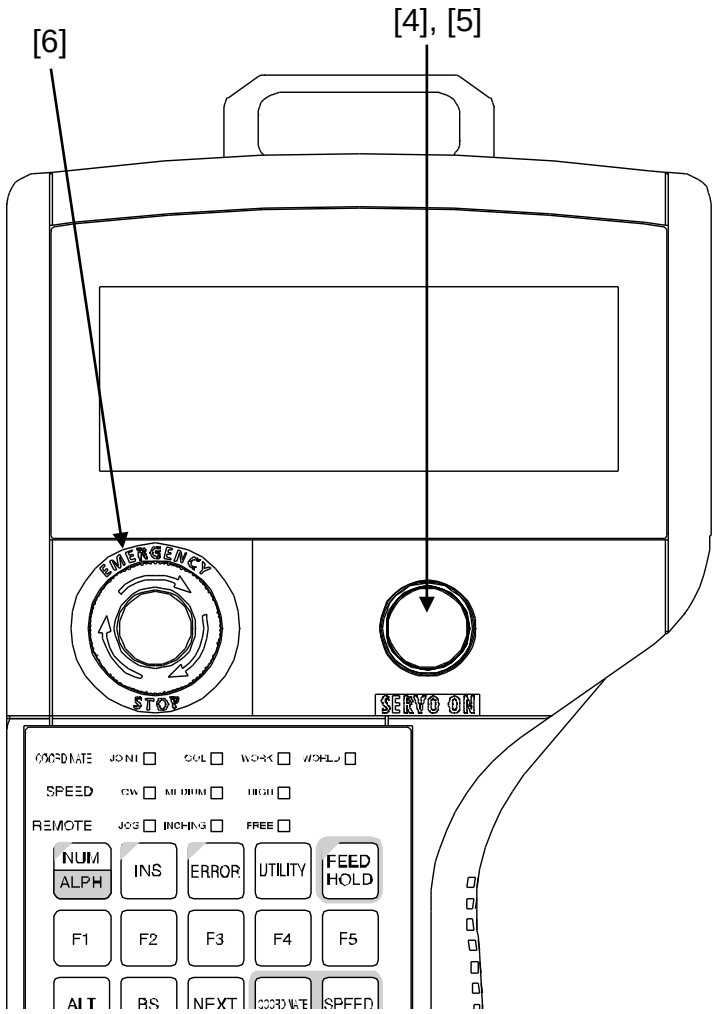


Fig. 3.6 EMERGENCY STOP switch provided on teach pendant

At this time, make sure that the EMERGENCY STOP switch [6] remains depressed.

3) Turn the EMERGENCY STOP switch [6] clockwise and make sure that the same switch turns off.

c) Make sure that the safety devices controlled by the external operation input signals work properly.

1) Press the SERVO ON switch [1] equipped on the control panel to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal “emergency stop” (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

2) Set ON the safety devices connected with external operation input signal “low speed command”, and make sure that each signal functions properly.

3.5.3 Battery Replacement

The SRAM equipped on the X8HC printed board of the robot controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the memory data, resulting in the data being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

- 1–145 MAIN Battery alarm

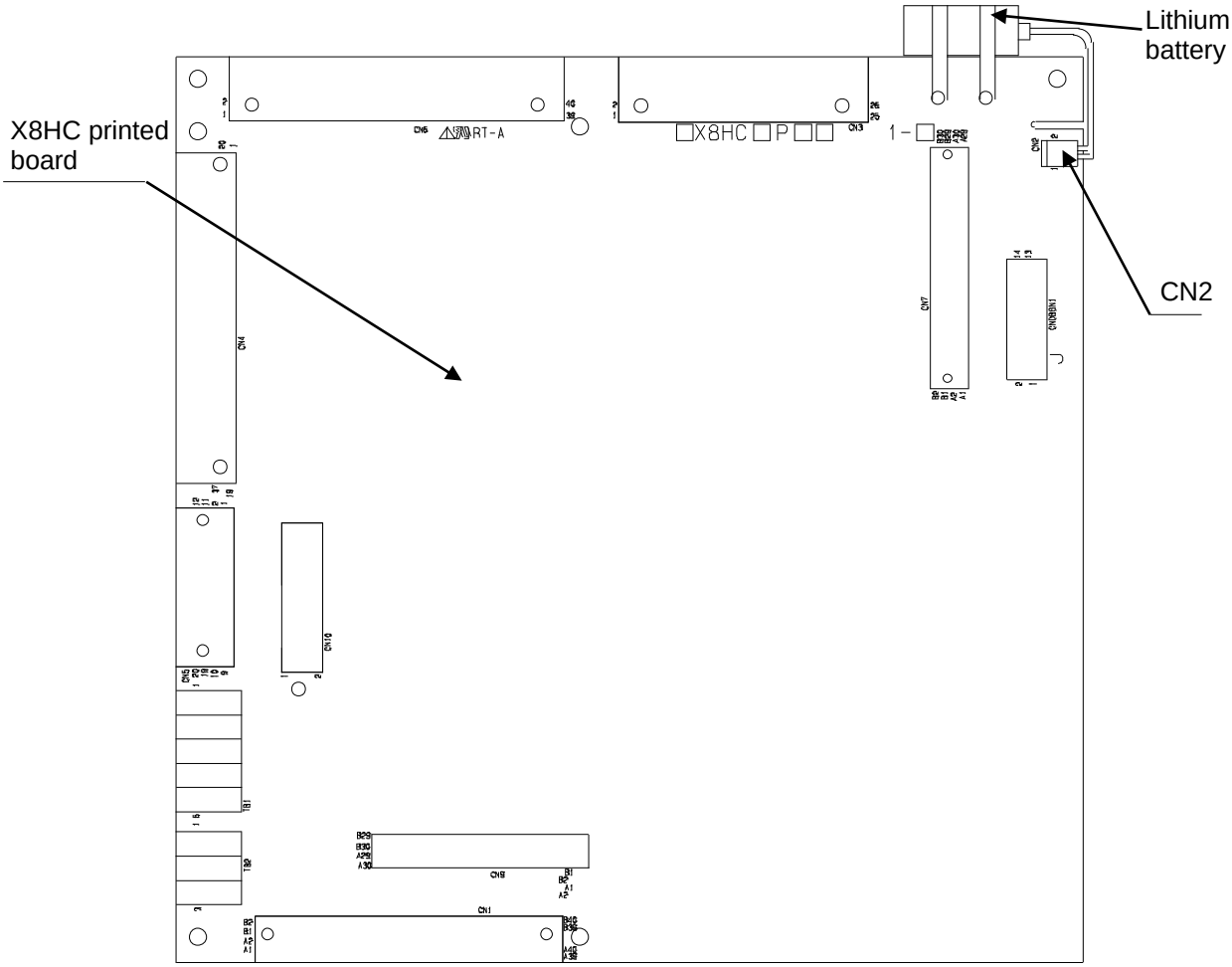


Fig. 3.7 Location of lithium battery

Table 3.1 Type of battery

Type of battery	Manufacturer	Battery service life
ER3V–C	Toshiba Battery	10 years (at room temperature)

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.

Only the battery shown in Table 3.1 above should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[Battery replacement procedures]

1) If the battery is to be kept removed for more than five (5) minutes, copy all programs and various parameters stored in the SRAM to the personal computer. In the worst case, all memory data are lost.
2) Set OFF the main power switch.
3) Remove the cover from the controller.
4) Disconnect the battery connector from CN2 of the X8HC printed board, and remove the battery from the TY-RAP securing the battery.
(For the battery position, see Fig. 3.7.)
5) Set the new battery within five (5) minutes.
Even if not used, the lithium battery capacity decreases by natural discharge. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.
6) Mount the cover on the controller.
7) If the battery is kept removed for more than five (5) minutes, turn on the main power switch and load the programs from the floppy disk unit of the personal computer. If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

! CAUTION

- Waste battery should be disposed of according to the user's in-house regulations.
- NEVER drop the battery into fire. NEVER short-circuit, charge, disassemble or heat it. Otherwise, liquid leakage or rupture may be caused.

3.5.4 Replacement of Switching Power Supply Unit

The life of the switching power supply unit (DC5 V/DC24 V) used in the robot controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor, and if the load current is large, the running time is long and the ambient temperature is high, the life will reduce.
As the life varies largely with the user's operating conditions (5 ~ 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[PS1 replacement procedures]

1) Turn off the main power switch.
2) Turn off the power breaker.
3) Remove the cover from the controller.
4) Disconnect all connectors in the controller, which are connected to the X8HC, X8HS (two (2) pcs.) and X8HP printed boards.
   (For the location of each printed board, see Fig. 3.3.)
5) Disconnect the X8HC, X8HN (X8HI), X8HS (two (2) pcs.) and X8HP printed boards from the controller.
6) Disconnect all cables coming from the power module and the cable between the PS1 and mother board.
7) Remove the three (3) screws securing the controller and PWB set bracket.
   (For the PWB set bracket location, see Fig. 3.3.)

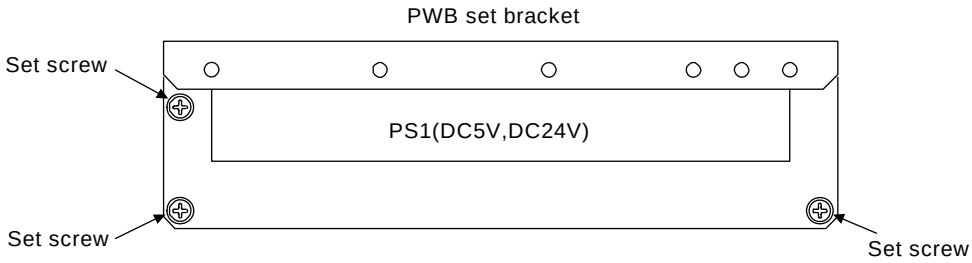


Fig. 3.8 PWB set bracket (upper side view)

8) While the PWB set bracket is tilted slightly, remove the two (2) connectors connected to the PS1, then disconnect the PWB set bracket from the controller.

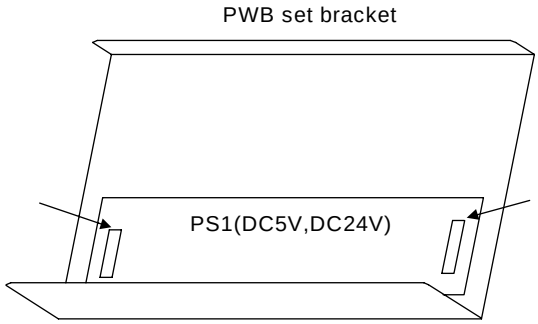


Fig. 3.9 PWB set bracket

9) Disconnect the old power supply from the PWB set bracket, then set the new power supply.

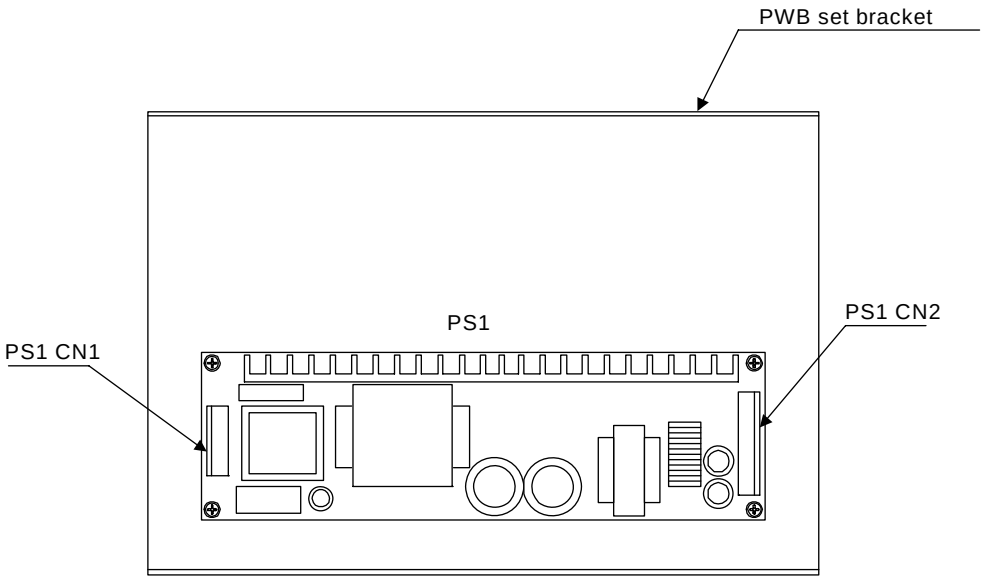


Fig. 3.10 PWB set bracket (front view)

10) Return the PWB set bracket mounting the new power supply to the controller, then connect the connectors.

11) Secure the PWB set bracket to the controller, using the three (3) screws. (See Fig. 3.8.)

12) Mount the disconnected printed boards (i.e., X8HC, X8HN (X8HI), X8HS (two (2) pcs.) and X8HP) to the controller.

13) Connect the disconnected connectors to respective printed boards.

14) Attach the cover to the controller.

[GS1 replacement procedures]

1) Turn off the main power switch.

2) Turn off the power breaker.

3) Loosen the terminal screws, using a screwdriver, disconnect the old power supply unit, then set the new power supply unit.
The location of the power supply unit is shown in Fig. 3.2 above.

3.5.5 Replacement of Fuse (X8HN, X8HI Printed Board)

If the current exceeding the specified current has run through the I/O unit, the fuse of the X8HN (X8HI) printed board is blown out. If the alarm saying “I/O Fuse broken (8–273)” is displayed, replace the fuse with a new one.
If the fuse of the output module has been blown out, examine and identify an fault circuit, remedy the cause, then replace the fuse.

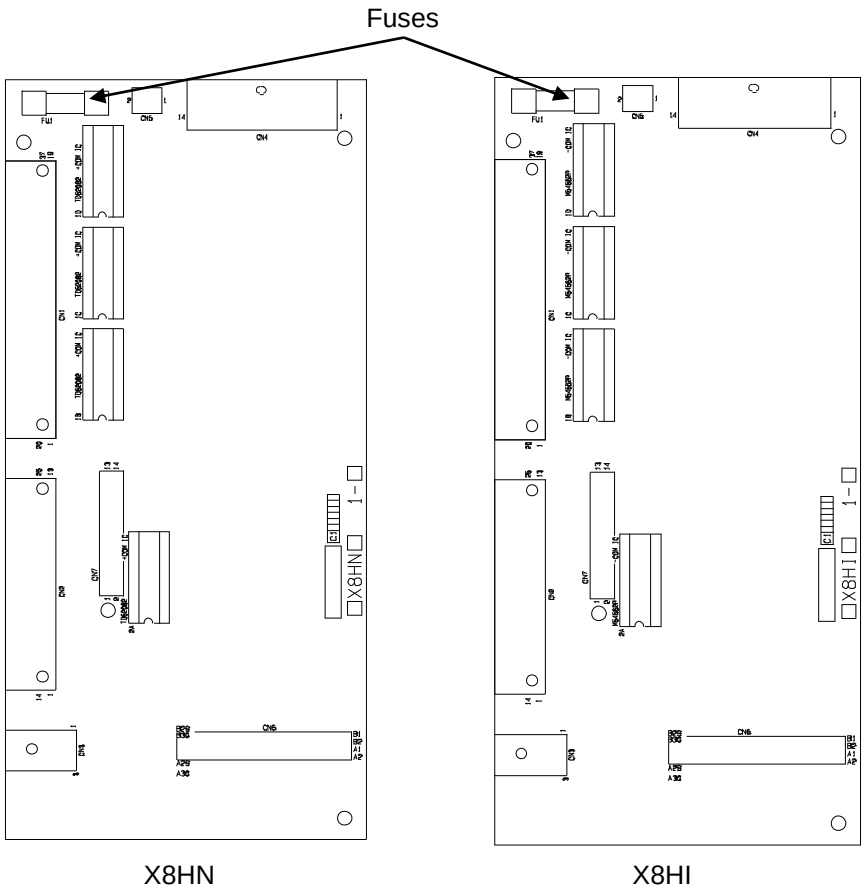


Table 3.2 Type of fuse

Type of fuse	Manufacturer
51NM030H	PICO

[Replacement procedures]

1) Turn off the controller main power.
2) Remove the cover attached to the fuse unit as shown above.
3) Remove the fuse and set a new fuse.
4) Mount the removed cover.
5) Turn on the controller main power and make sure that no error will occur.

3.5.6 Replacement of Output ICs (X8HN, X8HI Printed Board)

If the current exceeding the specified current has run through the output unit, output signals may not turn on normally. When this happens, the ICs on the X8HN (X8HI) printed board has possibly been damaged. Replace the ICs then.
Before the replacement, examine and identify an fault circuit, remedy the cause, then replace the ICs.

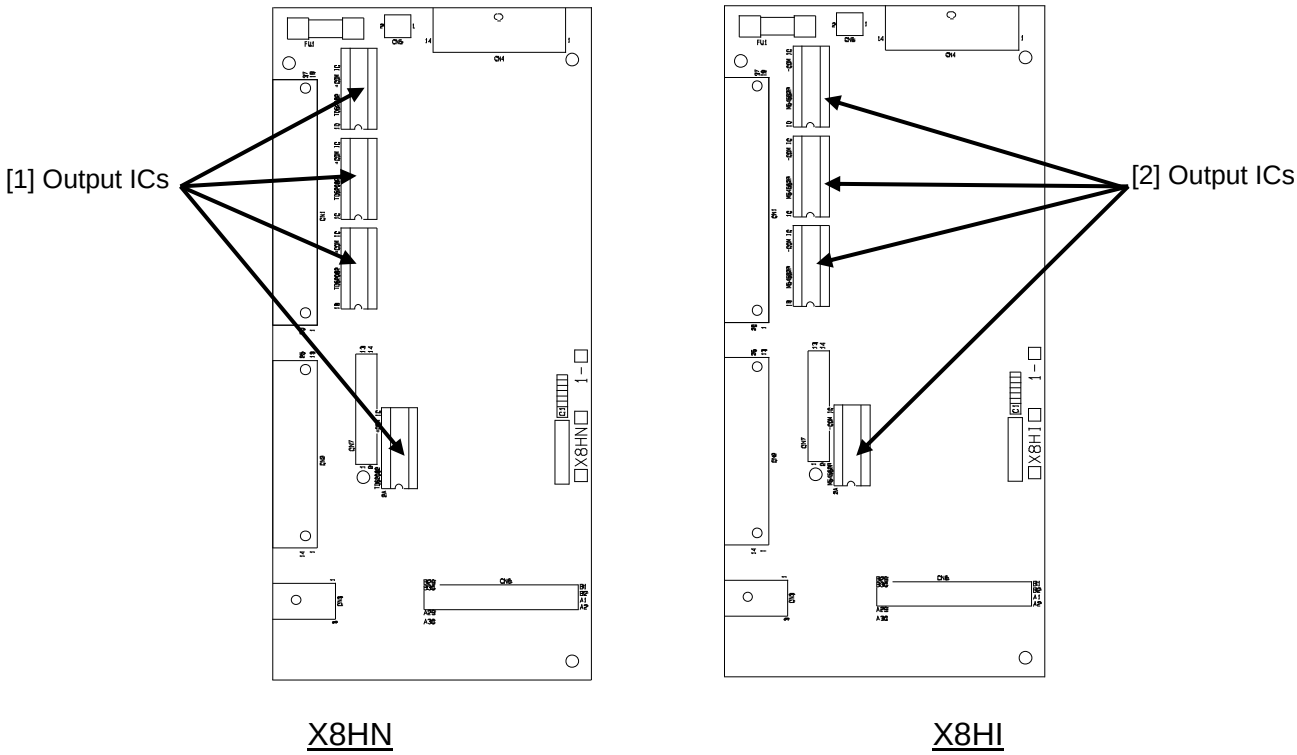


Table 3.3 Type of IC

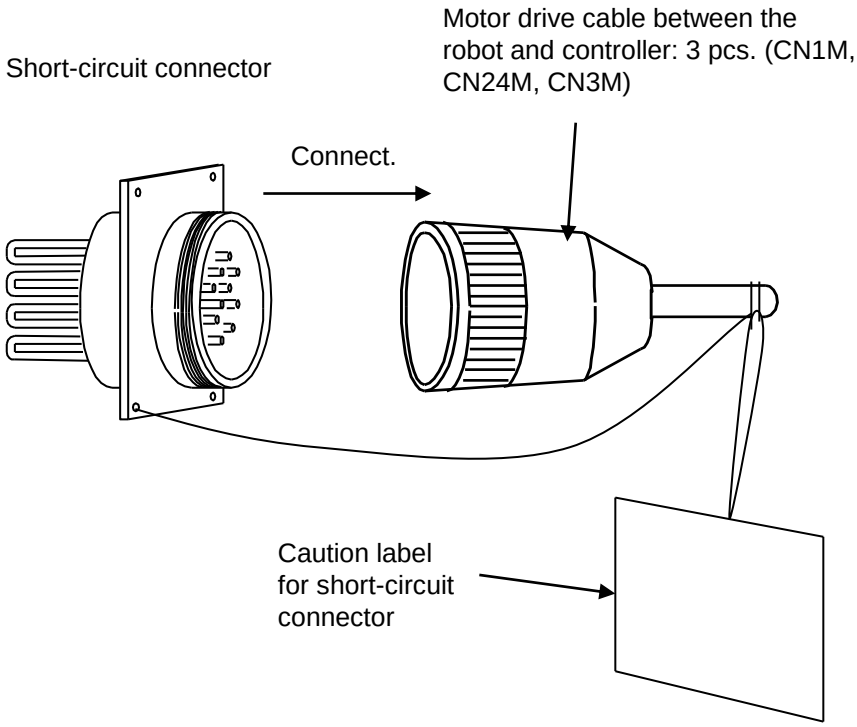
	Name of board	Type of IC	Manufacturer
[1]	X8HN	TD62082AP	Toshiba Corp.
[2]	X8HI	MM5456P	Mitsubishi Electric

[Replacement procedures]

1) Turn off the controller main power.
2) Remove the cover from the controller.
3) Disconnect all connectors connected to the X8HC printed board in the controller.
4) Disconnect the X8HC, X8HN (X8HI) printed boards from the controller.
5) Remove the ICs shown above from respective sockets.
6) Attach the new ICs to the sockets.
7) Mount the X8HC, X8HN (X8HI) printed boards on the controller.
8) Connect all disconnected connectors to the X8HC printed board, then attach the controller cover.
9) Turn on the controller main power.

3.5.7 Precautions on Transport

When the motor drive cables of the robot have been disconnected from the controller, be sure to attach the following short-circuit connectors (accessories) to them.
If the robot has been moved suddenly or heavy vibration has been imposed on the robot during power OFF without connecting the short-circuit connectors, the absolute coordinate counter of each encoder may not trace the robot position, resulting in generation of a “position detector (encoder) error” or home position data being lost.



Appendix Table 1 Parts List for Maintenance and Replacement: SR–1504HZ Main Robot

Part name	Type		Maker	Q'ty	Remarks
AC servo motor	MFM252C3GA	Y610A5410+Y5L023930	SHIBAURA MACHINE	1	Axis 1
	MDM152C2GA	Y610A5420+Y5L023930		1	Axis 2
	MDM302C3HA	Y610A5430+Y5L023930		1	Axis 3
	VLBST–Z06030S –0U001	Y610A1D90		1	Axis 4
Reduction gear	RV–160E–101	Y610A5440+Y610A3860	SHIBAURA MACHINE	1	Axis 1
	RV–80E–81	Y610A5450+Y610A3860		1	Axis 2
	RV–20E–81	Y610A5460+Y610A3860		1	Axis 4
Timing belt	656–8YU–40 (*1)		SHIBAURA MACHINE	1	Axis 3
	632–EV8YU–40 (*2) (Standardly equipped for machines delivered in July, 2007 and after.)			1	Axis 3
Timing pulley	S655000 (*1)		SHIBAURA MACHINE	1	Axis 3 drive side
	S816737 (*2) (Standardly equipped for machines delivered in July, 2007 and after.)			1	Axis 3 drive side
	S647499			1	Axis 3 driven side
Ball screw	L989628 (for Z = 600)		SHIBAURA MACHINE	1	Axis 3
	M208823 (for Z = 1000)				
	M013083 (for Z = 1300)				
	M198709 (for Z = 1500)				
Linear guide	L990441 (for Z = 600)		SHIBAURA MACHINE	2	Axis 3
	M208824 (for Z = 1000)				
	M013082 (for Z = 1300)				
	M198711 (for Z = 1500)				
Bearing	6020ZZ		SHIBAURA MACHINE	1	Axis 2
	6026ZZ			1	Axis 1
	7206–5GL–P5 (Koyo)			1	Axis 3
	BF30 Support unit (THK)			1	Axis 3
Grease	Alvania EPR0		Shell Oil	16 kg	
	Alvania grease S2			400 g	
Encoder backup battery	Size AA alkali battery			3	Battery box

*1: Used for the machines delivered before July, 2007.

*2: Used for the machines delivered in July, 2007 and after.

! CAUTION

- With the exception of the encoder backup batteries and grease, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- Replacement of the parts is performed by SHIBAURA MACHINE service engineer. If the customer changed the parts, we will not guarantee any subsequent troubles or accidents.

Appendix Table 2 Parts List for Maintenance and Replacement: Main Robot Cables

Part name	Type	Maker	Q'ty	Remarks
Arm side cable	KL3109G01	SHIBAURA MACHINE	1	Axis 1 motor drive cable
	KL3108G01		1	Axis 2 motor drive cable
	KL3106G01		1	Axis 4 motor drive cable
	KH6385G01		1	Axis 1 motor encoder cable
	KH6387G01		1	Axis 2 motor encoder cable
	KH6390G01		1	Axis 4 motor encoder cable
	KH6397G01		1	Hand I/O cable
For Z600 Cable in column	G96989G01	SHIBAURA MACHINE	1	Axis 1 ~ 4 encoder cable
	M209216		1	Axis 1, 2 motor drive cable
	M209217		1	Axis 1, 2 motor drive cable
	KL3117G01		1	Axis 4 motor drive cable
	G96990G01		1	Hand I/O cable
	KL3110G01		1	Axis 3 motor drive/brake cable
For Z1000 Cable in column	G96989G03	SHIBAURA MACHINE	1	Axis 1 ~ 4 encoder cable
	M208879		1	Axis 1, 2 motor drive cable
	M208880		1	Axis 1, 2 motor drive cable
	KL3117G02		1	Axis 4 motor drive cable
	G96990G03		1	Hand I/O cable
	KL3110G03		1	Axis 3 motor drive/brake cable
For Z1300 Cable in column	G96989G02	SHIBAURA MACHINE	1	Axis 1 ~ 4 encoder cable
	M209214		1	Axis 1, 2 motor drive cable
	M209215		1	Axis 1, 2 motor drive cable
	KL3117G04		1	Axis 4 motor drive cable
	G96990G02		1	Hand I/O cable
	KL3110G02		1	Axis 3 motor drive/brake cable
For Z1500 Cable in column	G96989G04	SHIBAURA MACHINE	1	Axis 1 ~ 4 encoder cable
	M198743		1	Axis 1, 2 motor drive cable
	M198744		1	Axis 1, 2 motor drive cable
	KL3117G03		1	Axis 4 motor drive cable
	G96990G04		1	Hand I/O cable
	KL3110G04		1	Axis 3 motor drive/brake cable

Appendix Table 3 Parts List for Maintenance and Replacement: Cables between Robot and Controller

Part name	Type	Maker	Q'ty	Remarks
7 m cable	KL3111G05	SHIBAURA MACHINE	1	CN24M: Axis 2, 4 motor drive cable
	KL3112G05		1	CN1M: Axis 1 motor drive cable
	KL3113G05		1	CN3M: Axis 3 motor drive cable
	KL3114G05		1	CN4: I/O cable
	H07475G05		1	CN3: Encoder cable
10 m cable	KL3111G04	SHIBAURA MACHINE	1	CN24M: Axis 2, 4 motor drive cable
	KL3112G04		1	CN1M: Axis 1 motor drive cable
	KL3113G04		1	CN3M: Axis 3 motor drive cable
	KL3114G04		1	CN4: I/O cable
	H07475G04		1	CN3: Encoder cable
15 m cable	KL3111G03	SHIBAURA MACHINE	1	CN24M: Axis 2, 4 motor drive cable
	KL3112G03		1	CN1M: Axis 1 motor drive cable
	KL3113G03		1	CN3M: Axis 3 motor drive cable
	KL3114G03		1	CN4: I/O cable
	H07475G03		1	CN3: Encoder cable
20 m cable	KL3111G02	SHIBAURA MACHINE	1	CN24M: Axis 2, 4 motor drive cable
	KL3112G02		1	CN1M: Axis 1 motor drive cable
	KL3113G02		1	CN3M: Axis 3 motor drive cable
	KL3114G02		1	CN4: I/O cable
	H07475G02		1	CN3: Encoder cable

Appendix Table 4 Home Point Setting Tool List

Part name	Type	Maker	Q'ty	Remarks
Home point alignment angle	L990442	SHIBAURA MACHINE	1	For setting axis 3 home point
Locate pin	S655131		1	
Hexagon socket head cap screw	12×50		2	
Home point alignment angle	L990444	SHIBAURA MACHINE	1	For setting axis 1, 2, 4 home point
Locate pin	S655130		1	
Hexagon socket head cap screw	6×120		1	
Retaining ring	NETW8		1	
Pin	FPBD5–P7.95–L8–B6		2	
Pipe	S655192	SHIBAURA MACHINE	1	Grease discharge tool for axis 1, 2

Appendix Table 5 Parts List for Maintenance and Replacement: Controller

Part name	Type		Maker	Q'ty	Remarks
PS1 (Switching power supply)	LEB100F–0524		Cosel	1	
GS1 (Switching power supply)	S82J–10024D		OMRON	1	
Lithium battery	ER3V–C		Toshiba Battery	1	
Printed board	X8HCA (Main control)	Y610A2100	SHIBAURA MACHINE	1	
	X8HA (Analog servo)	Y610A2150		2	
	X8HPA (Servo power supply)	Y610A2340		1	
	X8HBA (Mother board)	Y610A2140		1	
	X8HNK (I/O board)	Y610A2120		1	Minus (–) common
	X8HJ (Connector)	Y610A2320		1	
	X8HO (Operation panel)	Y610A2310		1	
	HIC (Constant data board)	Y610A06A0		1	
Teach pendant	TP1000		SHIBAURA MACHINE	1	With 5 m-long cable
Operation panel unit	TS2000CP1 (X8HC)	Y610A2310	SHIBAURA MACHINE	1	
Connector panel	TS2000CM (X8HJ)	Y610A2320	SHIBAURA MACHINE	1	
System disk	TS2000SYS		SHIBAURA MACHINE	1	

Appendix Table 6 Servo Unit Application Table

Part name	Type		Maker	Q'ty	Remarks
AB28A	Servo unit	Y5L026250	SHIBAURA MACHINE	2	For axis 1, 3
AB14C	Servo unit	Y5L026130		1	For axis 2
AL06C	Servo unit	Y5L026370		1	600 W or less